

Bonsai: Exploratory analysis of single-cell data



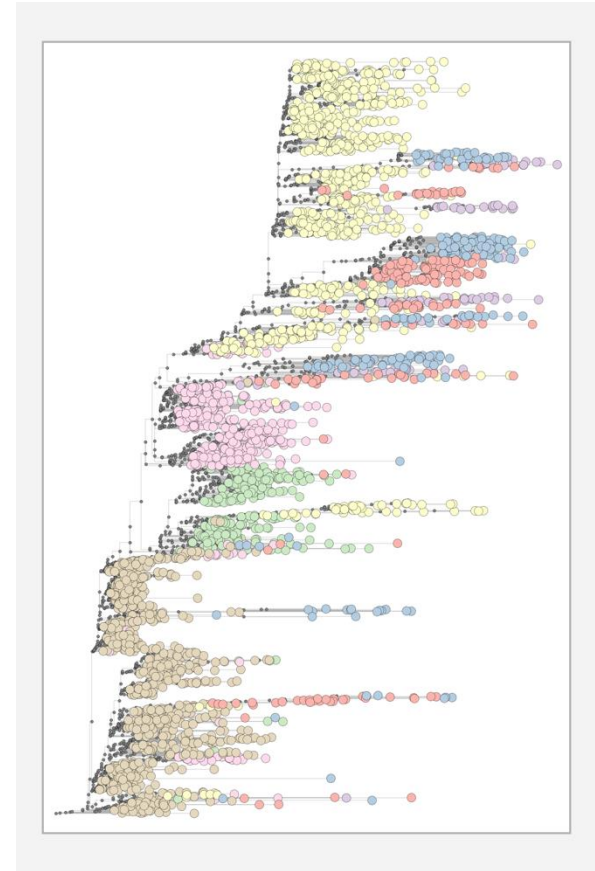
Sarah Morillo Leonardo
PhD student
Van Nimwegen lab



Daan de Groot
Research associate
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Mikhail Pachkov
Research Associate
Van Nimwegen lab



VIENNA BIOCENTER PHD PROGRAM



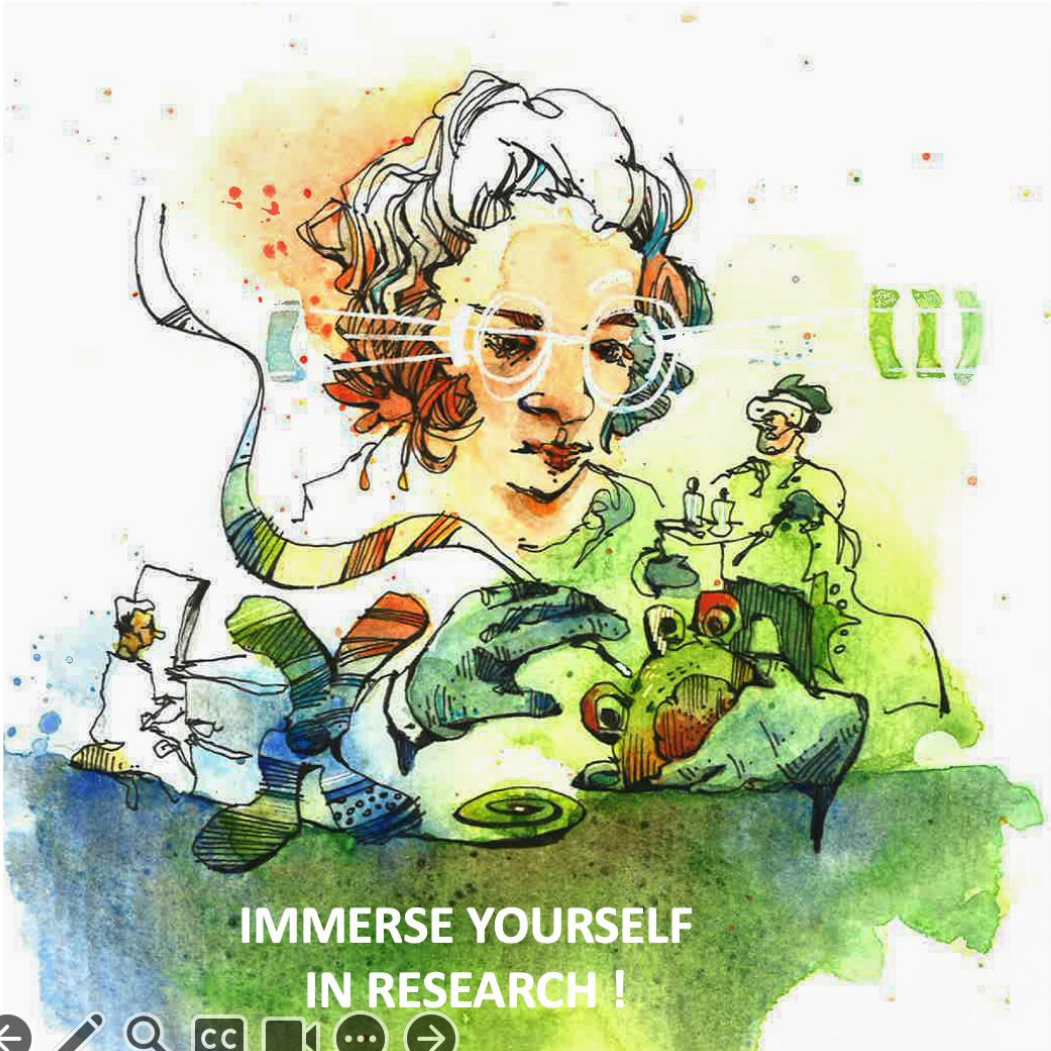
PhD Program in the Molecular Biosciences

- 4-year program
- Fully funded positions
- International interdisciplinary campus
- English as working language

Spring call open now! (deadline April 15, 2026)

Applicants eligible from all fields in the life sciences, including biology, biochemistry, plant science, computational biology, (bio)physics and many more.

More information: <https://training.vbc.ac.at/phd-program/>

A colorful, abstract illustration of a person with long, flowing hair and glasses, wearing a lab coat and gloves, working with laboratory equipment. The background is a mix of green and blue watercolor washes.

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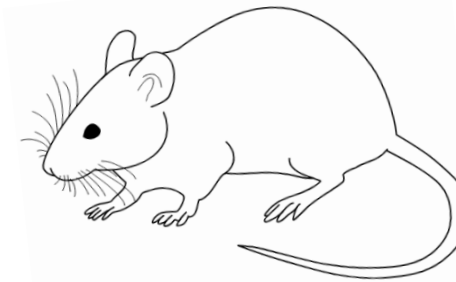
MEDIZINISCHE
UNIVERSITÄT WIEN

Today's program

- **14:00 – 14:30:** Challenges in scRNA-seq data analysis
- **14:30 – 15:00:** Our solution: *Sanity* + *Bonsai*
- **15:00 – 16:15:** Hands-on tutorial on running *Bonsai* and *Bonsai-scout*
- **16:20 – 16:50:** From dataset to paper figure
- **17:00 – 18:00:** Q&A + individual dataset analysis

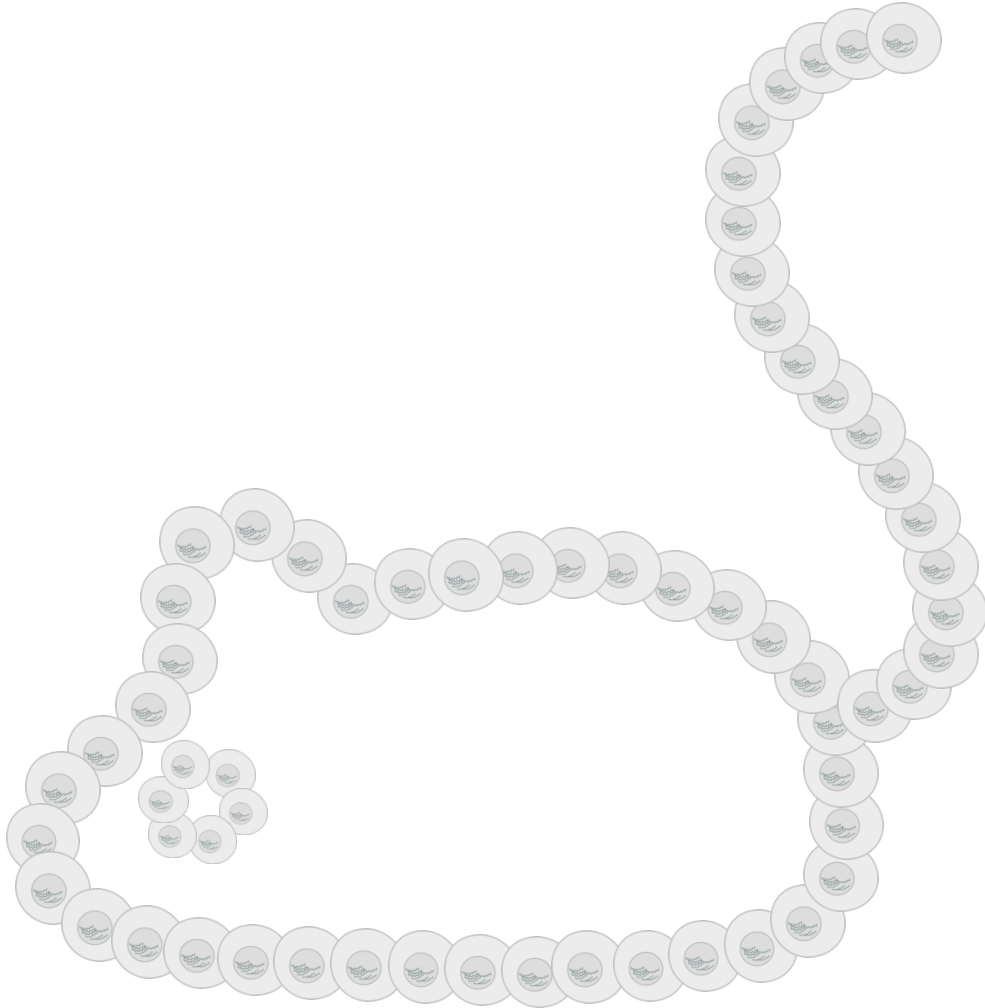
Theory and computational methods

How do single cells decide?



We can measure cellular states using **single-cell omics**

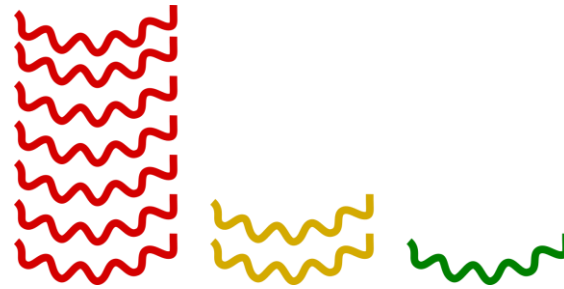
The promise of single-cell omics



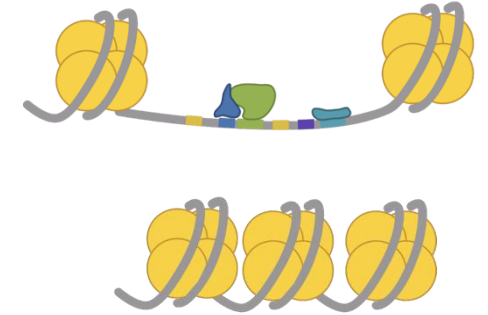
The promise of single-cell omics



scRNA-seq



scATAC-seq



The promise of single-cell omics



Typical scRNA-seq counts

20,000 features

	Cell 1	Cell 2	Cell 3	...
Gene 1	1	0	1	
Gene 2	0	0	0	
Gene 3	1	2	3	
Gene 4	0	0	0	
Gene 5	0	0	0	
Gene 6	8	5	9	
Gene 7	0	0	0	...

100 to 10,000,000 cells

The challenges of single-cell omics

1. The data are dominated by noise

Typical scRNA-seq counts

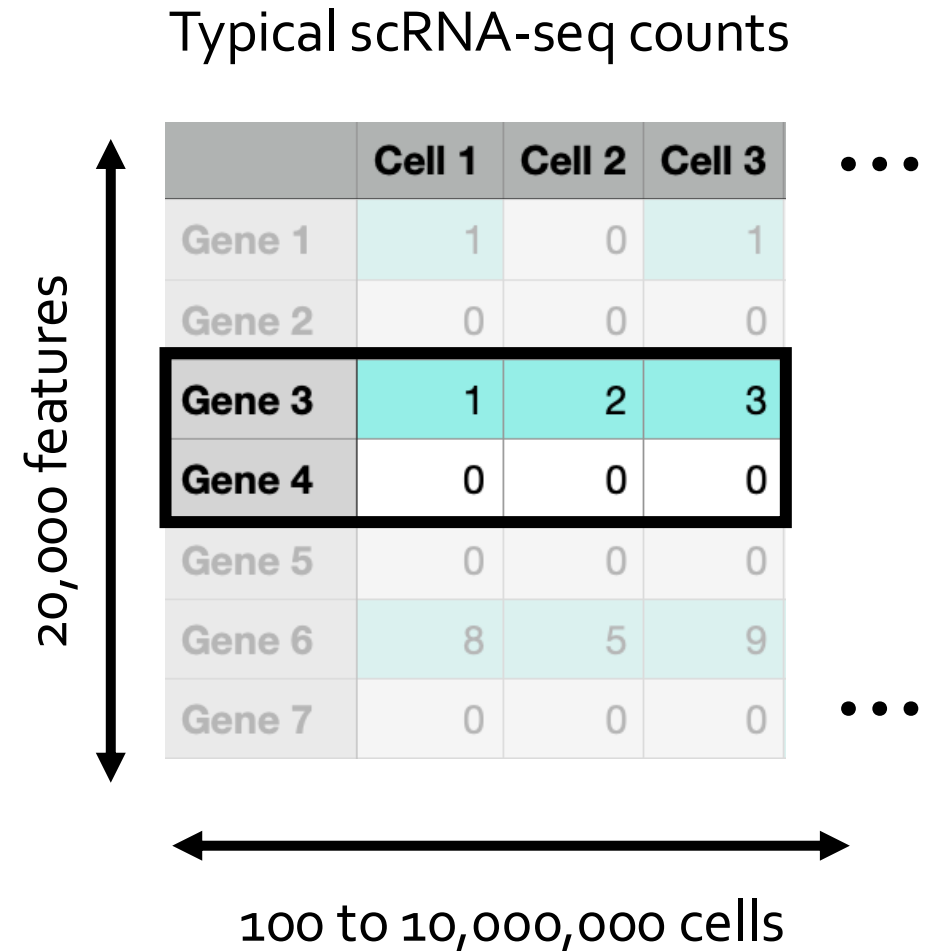
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The challenges of single-cell omics

1. The data are dominated by noise
2. We only have static snapshots



The challenges of single-cell omics

1. The data are dominated by noise

2. We only have static snapshots

3. Questions are often exploratory

How to explore >20,000 dimensions?

Typical scRNA-seq counts

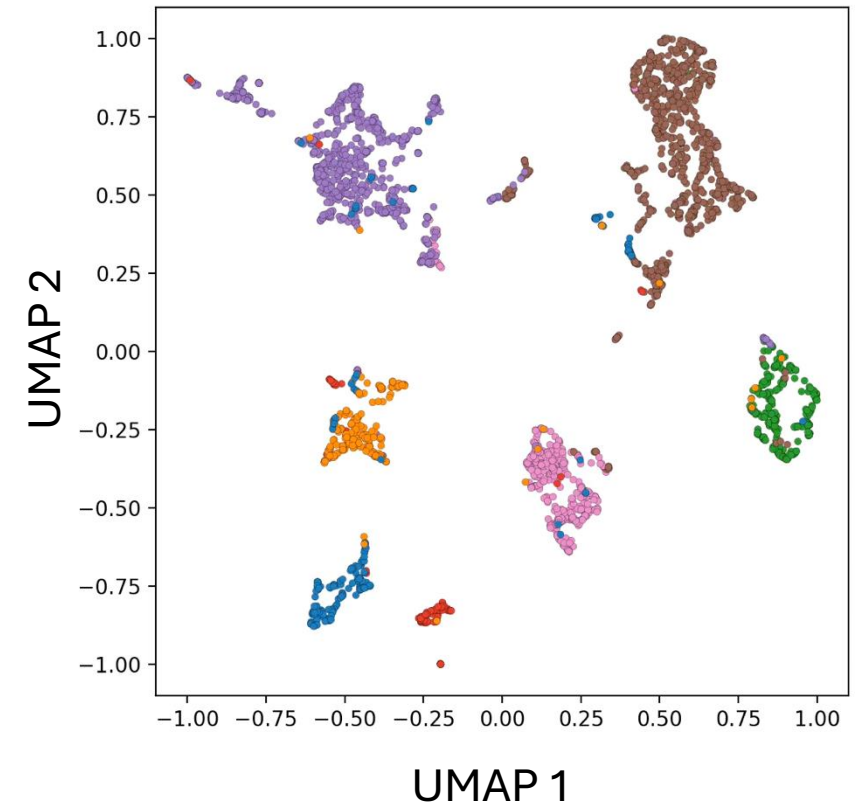
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How to explore >20,000 dimensions?

Current methods rely on 2D-embeddings



How to explore $>20,000$ dimensions?

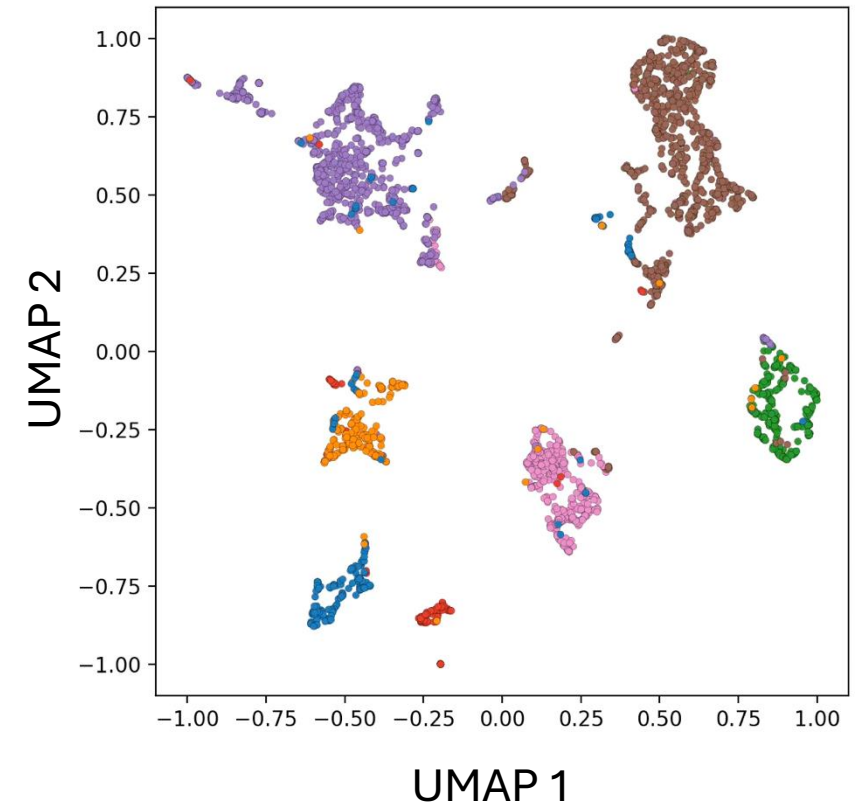
Current methods rely on 2D-embeddings

However, 2D-embeddings

- Cannot capture high-dimensional structures
- Require many arbitrary parameter choices
- Use coordinates that are uninterpretable

Still, they are used in many fields

- single-cell omics
- cytometry
- metagenomics
- population genetics
- neuronal spiking behavior
- image and text categorization

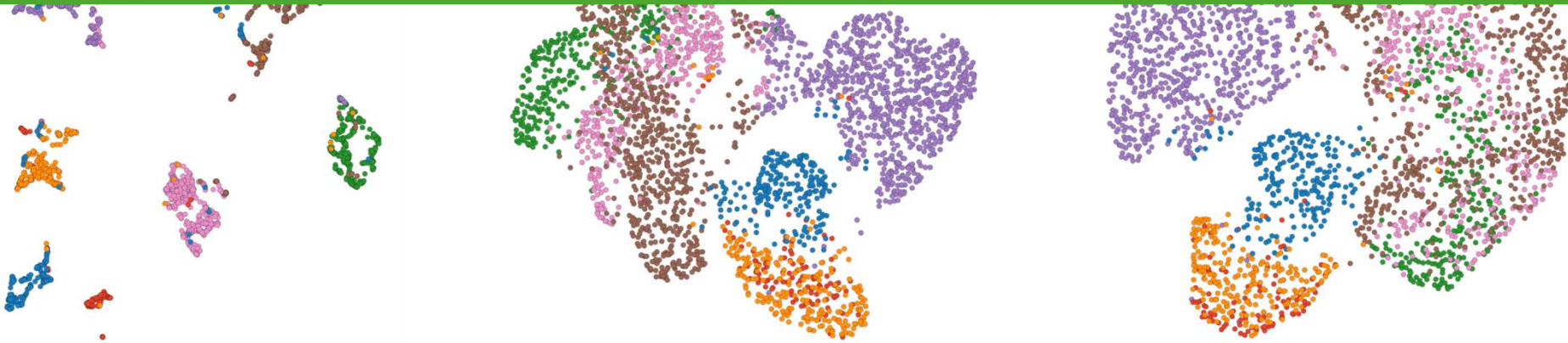


How to explore $>20,000$ dimensions?

Current methods rely on 2D-embeddings

“But there’s no better alternative!”

How to work with noisy, high-dimensional data?



Case study: How to analyze scRNA-seq data?

Case study: Why is scRNA-seq analysis hard?

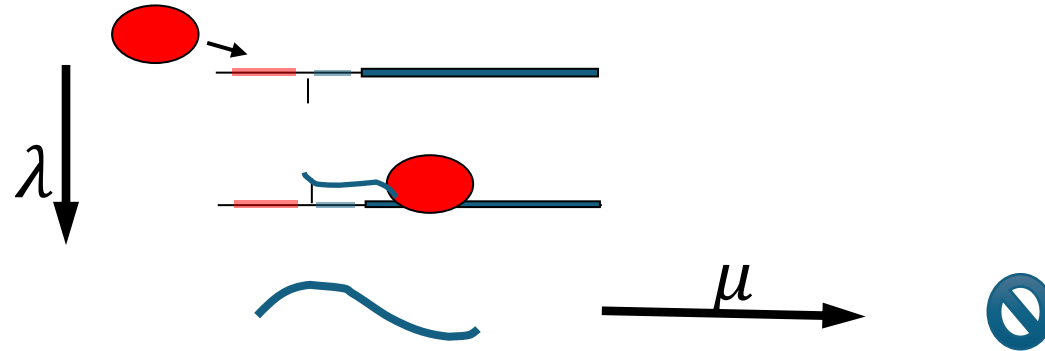
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1. Noisy: The number of captured mRNAs is affected by both measurement noise and intrinsic biological noise
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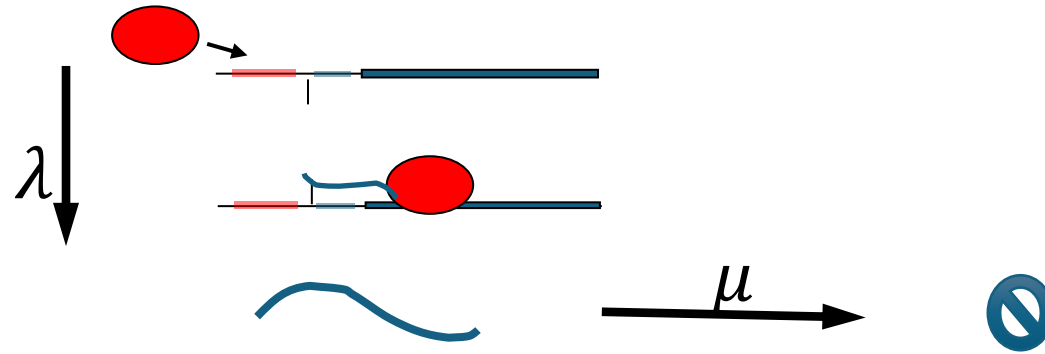
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Biological noise in scRNA-seq



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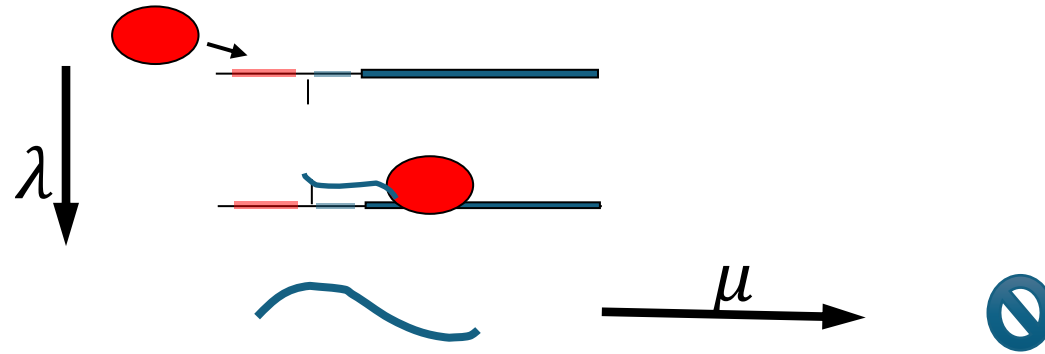
- Even at a constant transcription rate λ and degradation rate μ , the mRNA-numbers fluctuate

Transcriptional activity (expected mRNA count): $a = \frac{\lambda}{\mu} = \langle n \rangle$

Poisson distribution of mRNA number: $P_n = \frac{1}{n!} a^n e^{-a}$

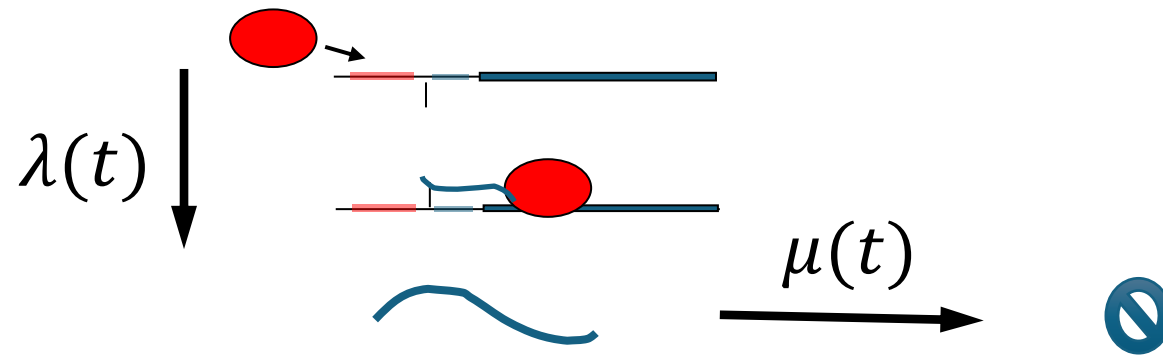
Variance equals mean: $\langle n \rangle = \text{var}(n) = a$

Biological noise in scRNA-seq



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- The state of the cell sets the **rates, rather than the mRNA numbers**

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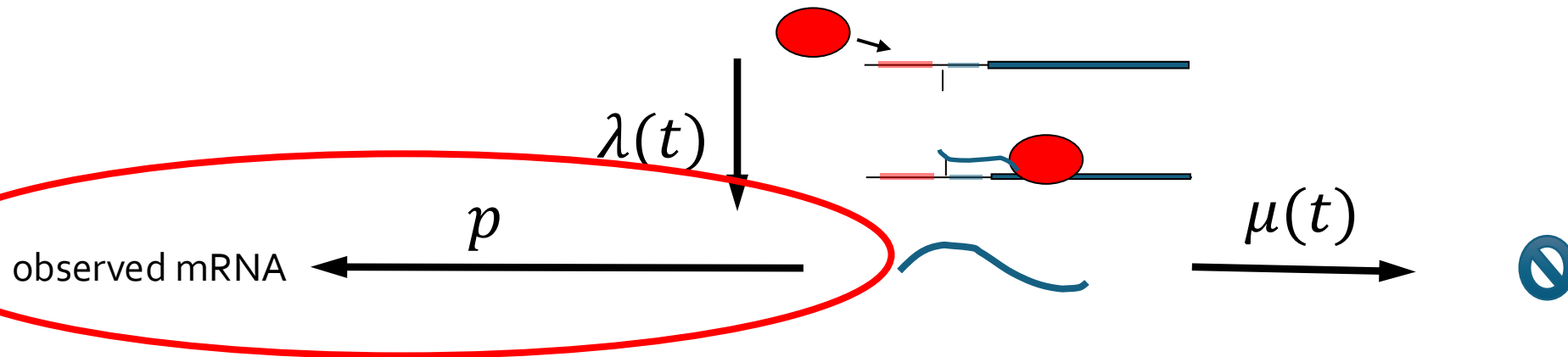
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Poisson distribution of mRNA number:
$$P(m_g | a_g) = \frac{1}{m_g!} (a_g)^{m_g} e^{-a_g}$$

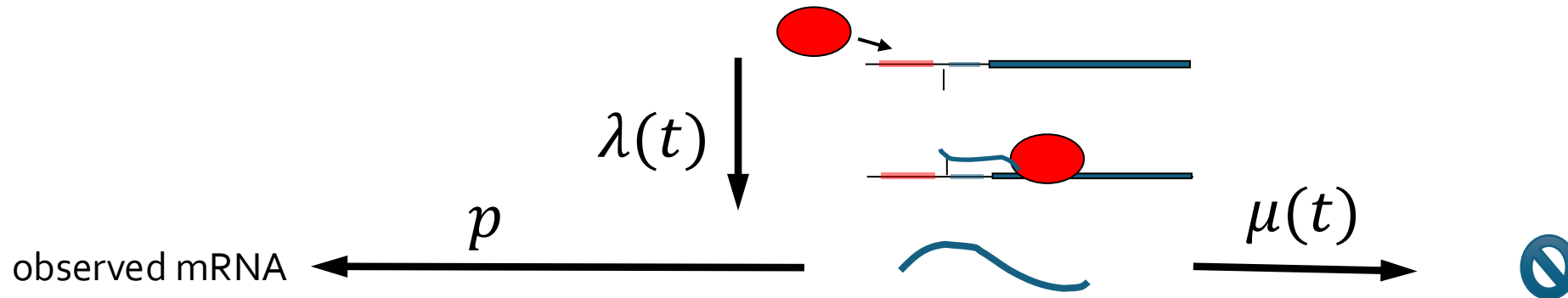
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Biological + technical noise in scRNA-seq



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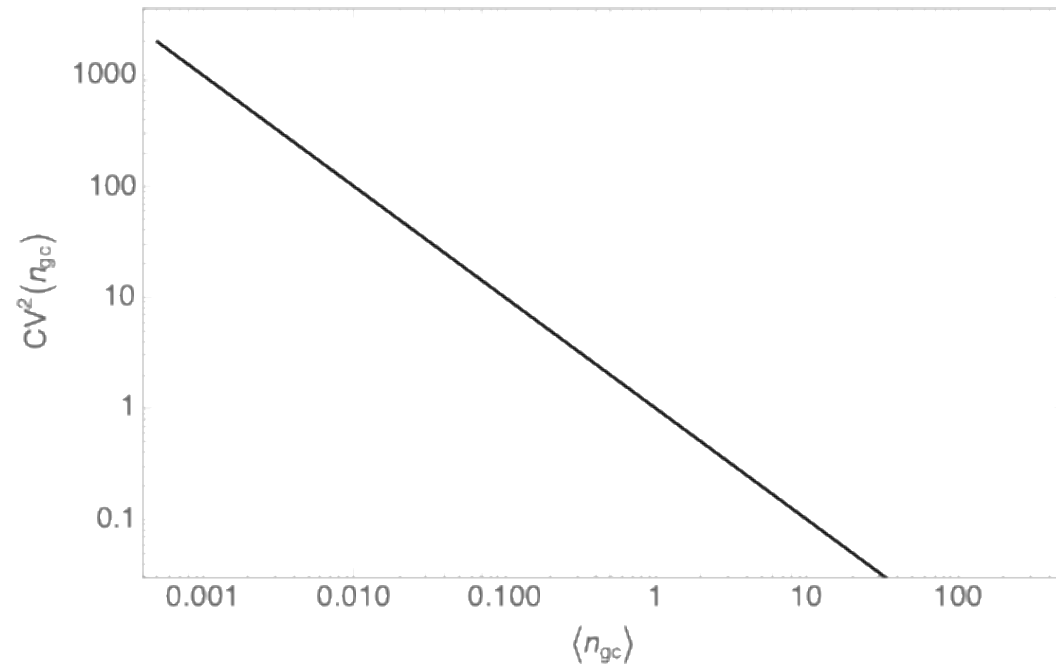
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Considering the underlying physics taught us:

- The noise in mRNA-counts should be Poisson
 - Prediction: Coefficient of variation should go down with expression level: $CV(n_g) \propto \frac{1}{\sqrt{n_g}}$
- Snapshots of mRNA-numbers can only yield information on “average” transcriptional activity a

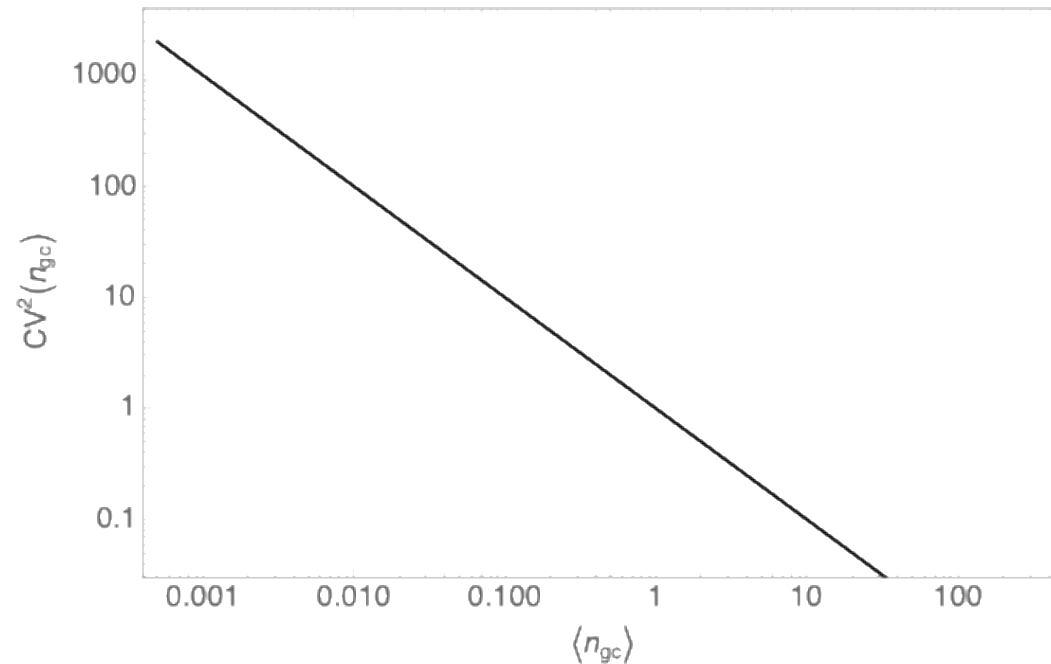
Testing the predicted noise statistics

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Power analysis of single-cell RNA-sequencing experiments

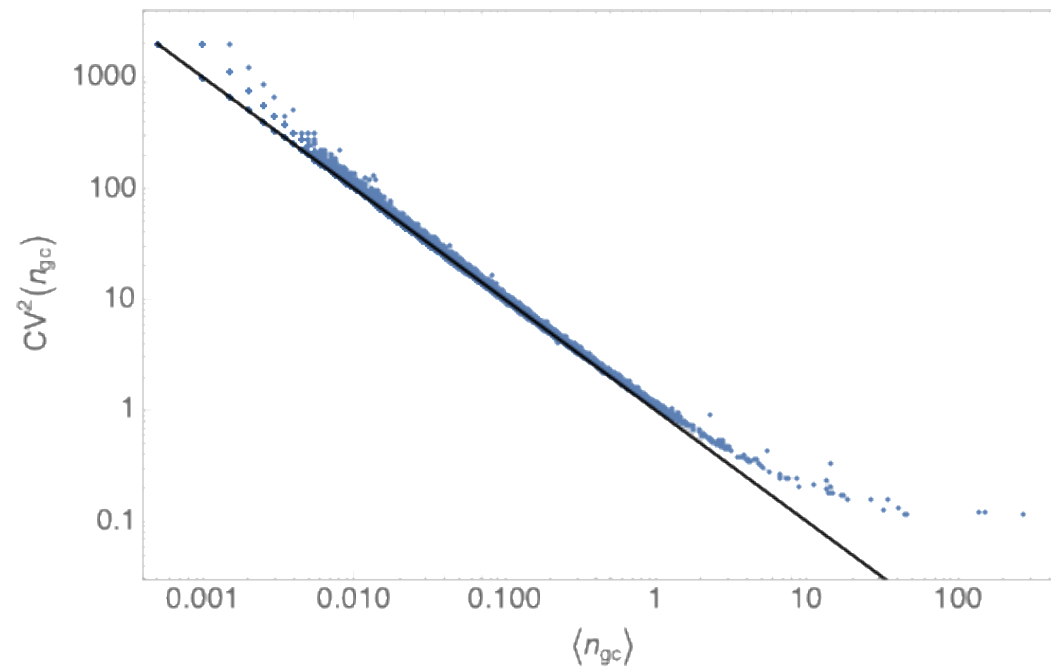
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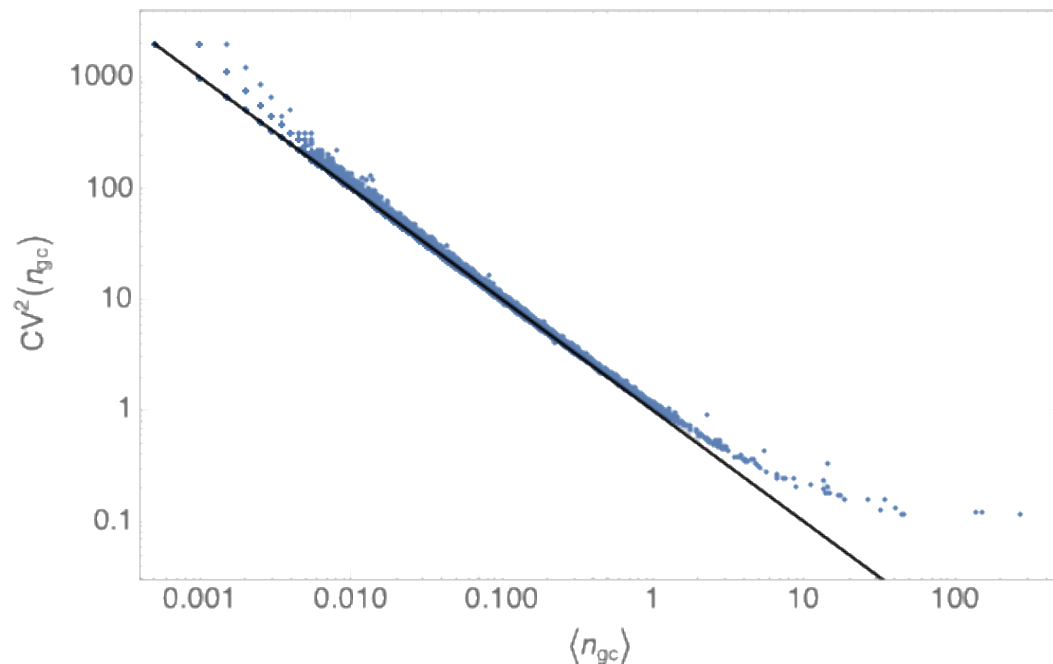
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- The CV^2 is clearly larger at high mean expression
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- **What is the origin of this overdispersion?**

Power analysis of single-cell RNA-sequencing experiments

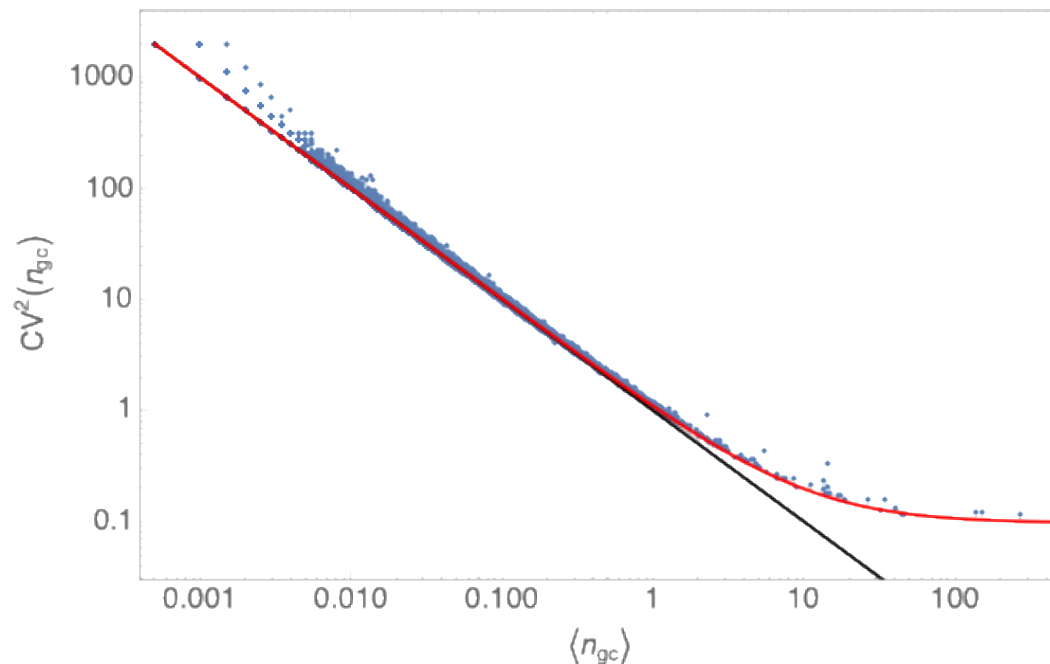
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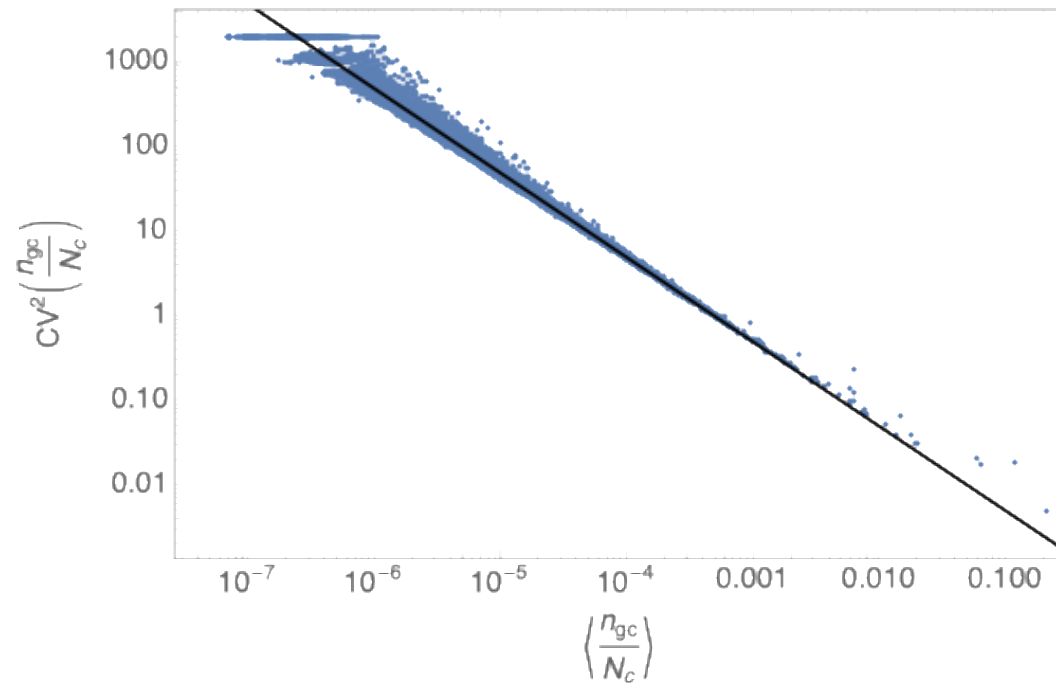
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 - But, the total mRNA-count N_c fluctuates from cell to cell
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 - Not the total count n_{gc} but the fraction n_{gc}/N_c should show pure Poisson fluctuations.

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Power analysis of single-cell RNA-sequencing experiments

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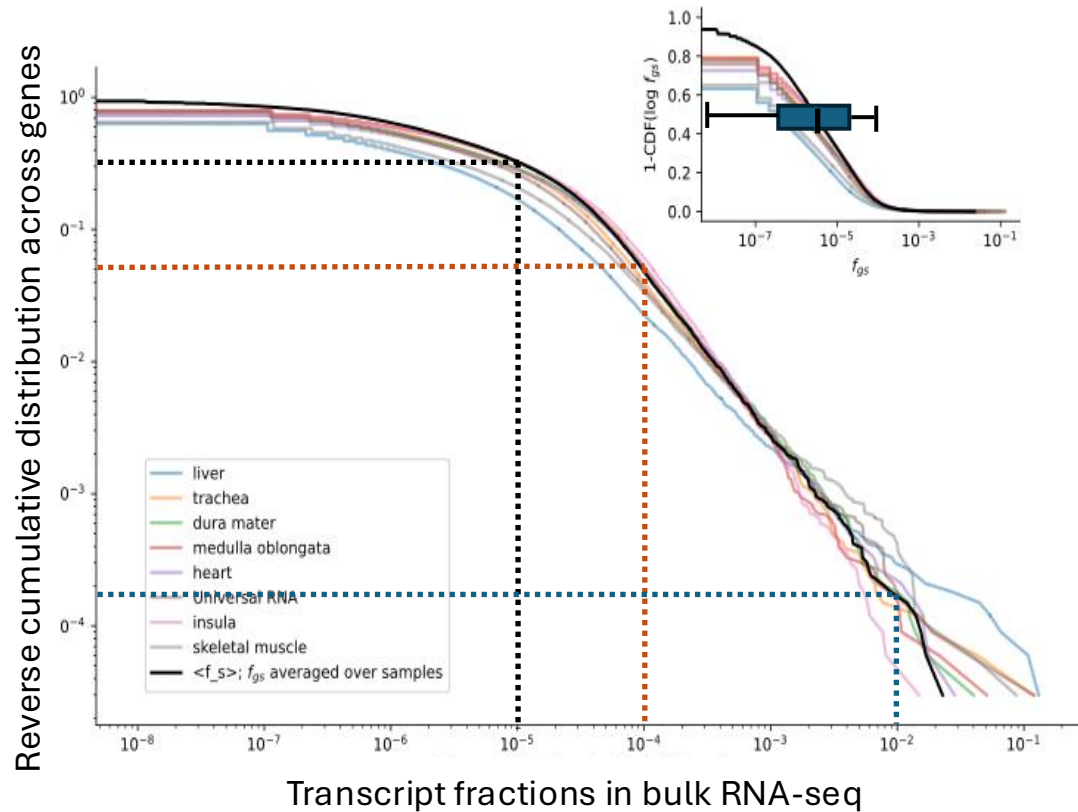
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How much do the expression levels vary?



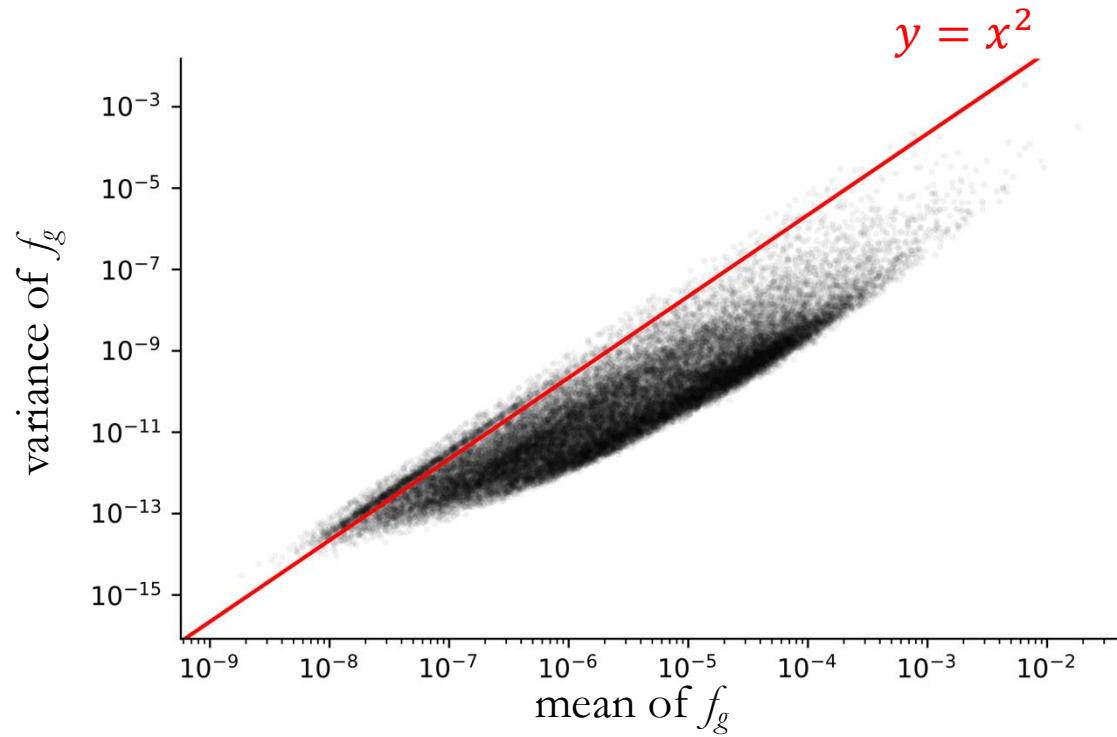
- 70% of the genes have less than 1 mRNA per 100'000 mRNAs.
- 95% of genes have less than 1 mRNA per 10'000 mRNAs.
- A few genes have more than 1 mRNA per 100 mRNAs.

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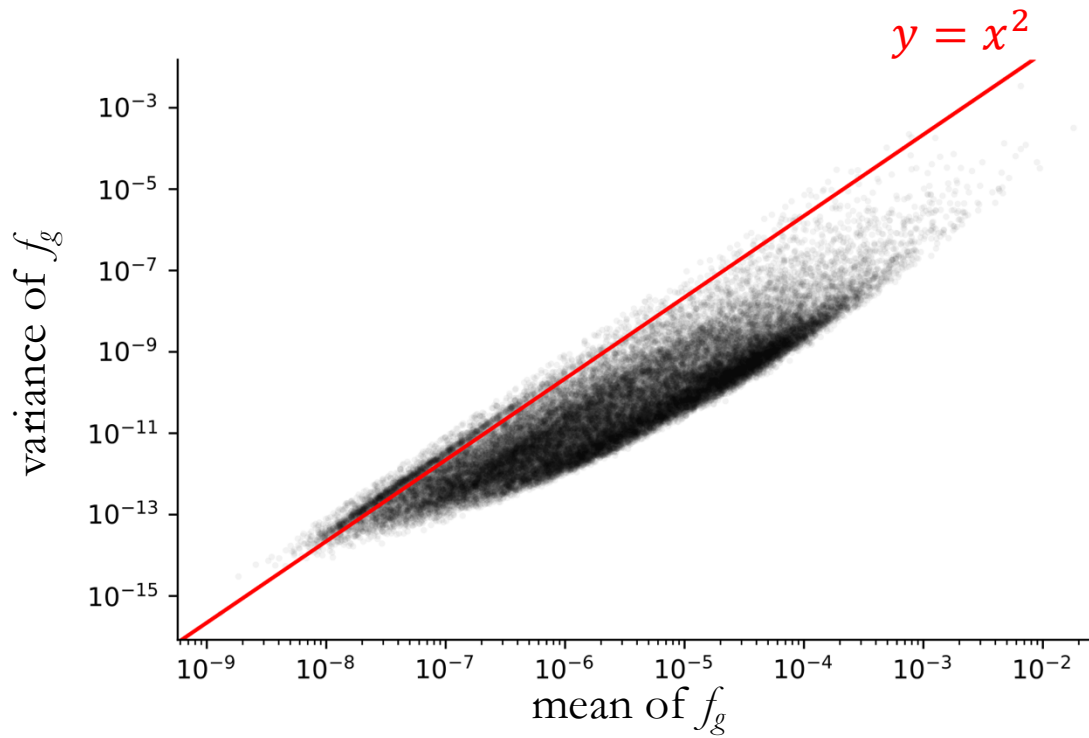
High-expressed genes vary more

- f_g = fraction of mRNAs that are mRNAs for gene g .
- Variance of f_g against mean of f_g across tissues/cell type.
- Expression changes are proportional to their mean.

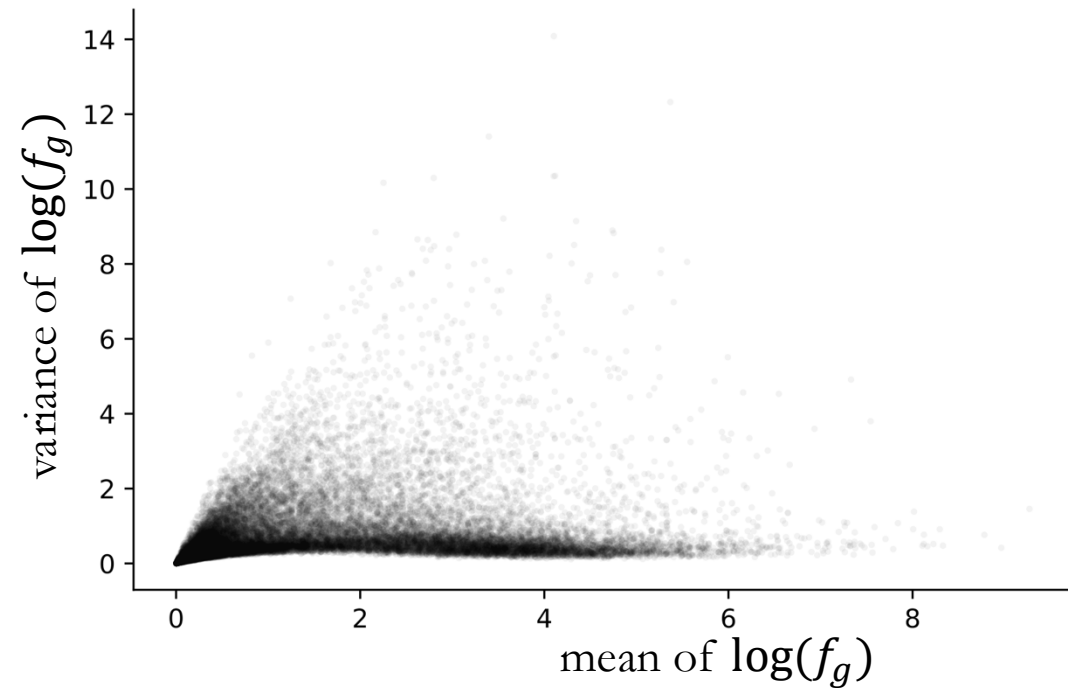


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Mean and variance of $\log(f_g)$ are independent.



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Case study: How to analyze scRNA-seq data?

1. The physical observable to determine is the “log transcription quotient”

- The cell's recent history of transcription and decay rates is summarized in the *transcription activities*:

$$a_{gc} = \int_0^\infty \lambda_{gc}(t) \exp \left[- \int_0^t \mu_{gc}(\tau) d\tau \right] dt$$

- To take variations in total mRNA count N_c into account, we look at *transcription quotients*: $q_{gc} = \frac{a_{gc}}{\sum_{g'} a_{g'c}} = \left\langle \frac{n_{gc}}{N_c} \right\rangle$.
- Because variation scales with mean expression, we estimate *log transcription quotients*: $x_{gc} = \log(q_{gc}) = \log\left(\frac{a_{gc}}{\sum_{g'} a_{g'c}}\right)$

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How to estimate these “LTQs”
from the mRNA-counts?

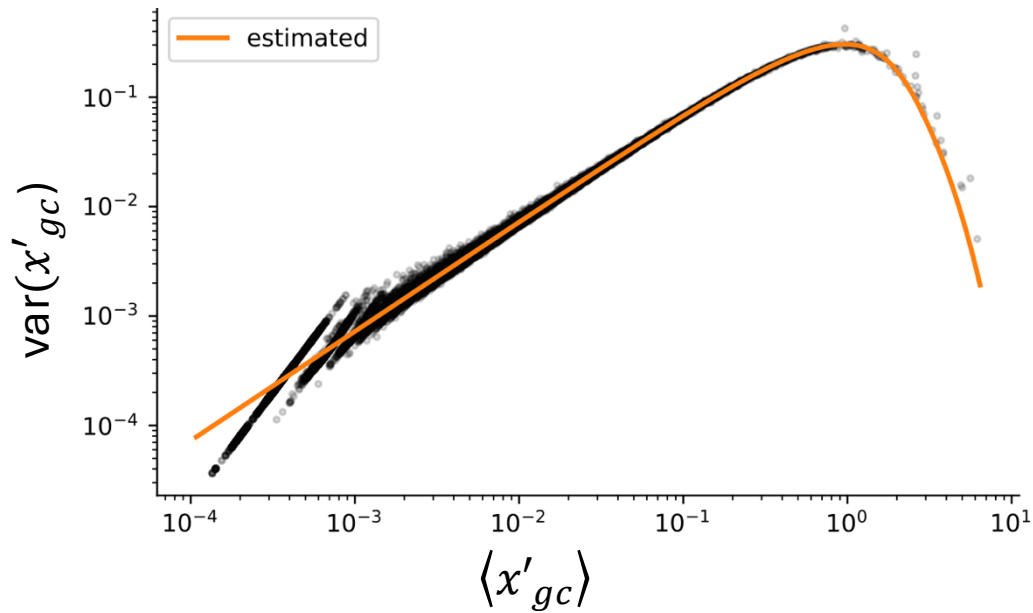
A popular option: **logp1-normalization**

How to estimate these “LTQs”
from the mRNA-counts?

- Note that: $\langle n_{gc} \rangle = N_c e^{x_{gc}}$
- This suggests we may estimate x_{gc} just as: $x_{gc} = \log\left(\frac{n_{gc}}{N_c}\right)$ but this is not defined when $n_{gc} = 0$.
- Ad hoc fix: Add an offset and rescale by the average (or median) total UMI count $N = \langle N_c \rangle$: $x'_{gc} = \log\left(N \frac{n_{gc}}{N_c} + 1\right)$
- But the actual count n_{gc} differs from the expectation $\langle n_{gc} \rangle$ by Poisson noise!

A popular option: **logp1-normalization**

Aliquots from a homogeneous mRNA mixture



Power analysis of single-cell RNA-sequencing experiments

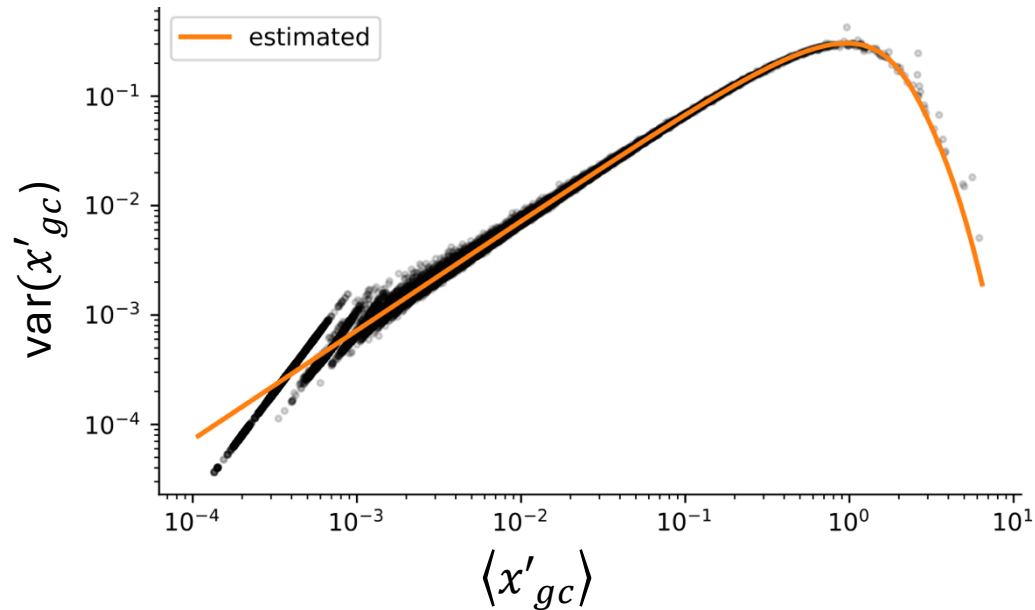
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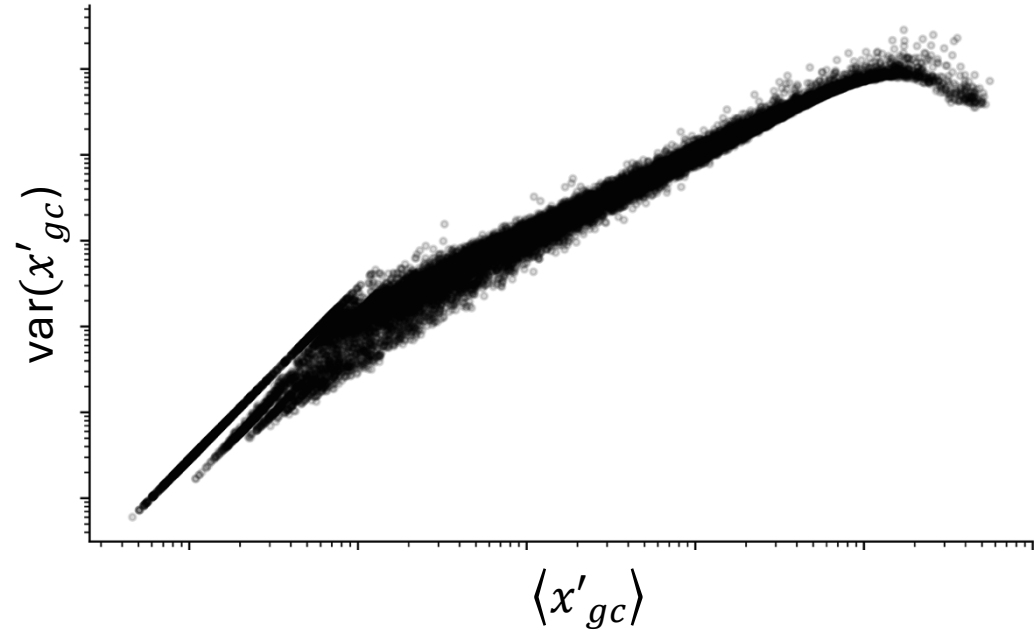
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- **Note: The true variance is zero for all genes**

A popular option: **logp1**-normalization

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Data from real cells



- Normalization with “logp1” does not correct for the dominant Poisson sampling noise
- To deal with this noise, most pipelines therefore follow-up with dimensional reduction

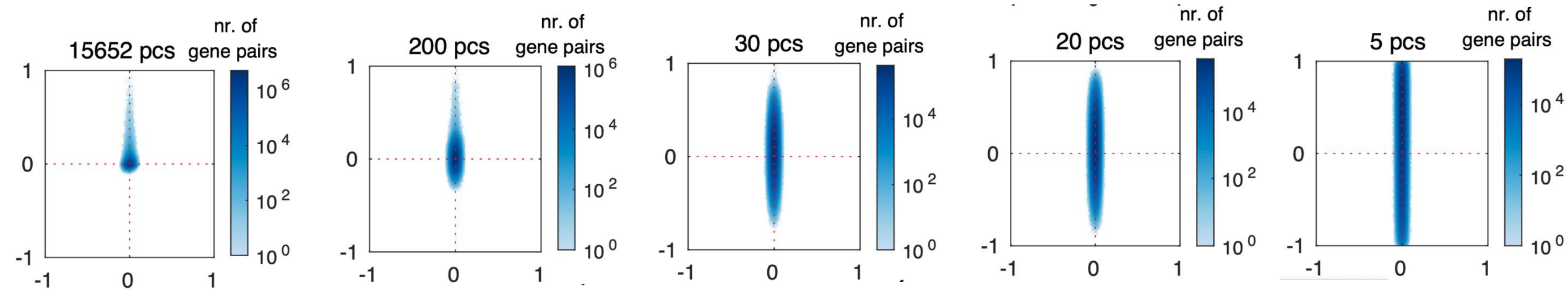
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 4. **Removing noise by dimensionality reduction can severely distort the data.**
2. High-dimensional: Faithfully representing the structure of high dimensional data is difficult
3. Hard to visualize: Current visualization methods are unreliable and uninterpretable.

Dimensionality reduction can introduce artifacts

- Simulated dataset with low true correlation between genes
- Dimension reduction may reduce the effect of sampling noise, but induces artifactual correlations

Taking logp1 normalization and projecting on the first n PCA components



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2. **High-dimensional: Faithfully representing the structure of high dimensional data is difficult**
 1. **kNN graphs remove much structure in the data.**
3. Hard to visualize: Current visualization methods are unreliable and uninterpretable.

Problems with the kNN-graph

What is a kNN graph?

1. Determine the distances between all pairs of cells.
2. For each cell, find the k neighbors that are closest to it.
3. Make a graph by drawing edges from each cell to its k nearest neighbors.

The omnipresence of the kNN-graph

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The use of kNN graphs to represent structure in scRNA-seq

- Averaging gene expression over nearest-neighbors.
- **Pseudotime** and **trajectory** analysis find trajectories on the kNN-graph.
- *Louvain* and *Leiden* **clustering** algorithms identify 'communities' in these graphs.
- **Visualization**: t-SNE and UMAP try to preserve the kNN graph in the embedding.

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Questions

- Is a kNN graph a good summary of the structure in the data?
- How accurately can we find nearest neighbors given that the data is so noisy?

Problems with the kNN-graph

Example 3-nn graph



- For some nodes the 3rd neighbor is very nearby.
- For others even the first neighbor is very far.
- This difference is *ignored* in kNN-graphs.

Problems with the kNN-graph

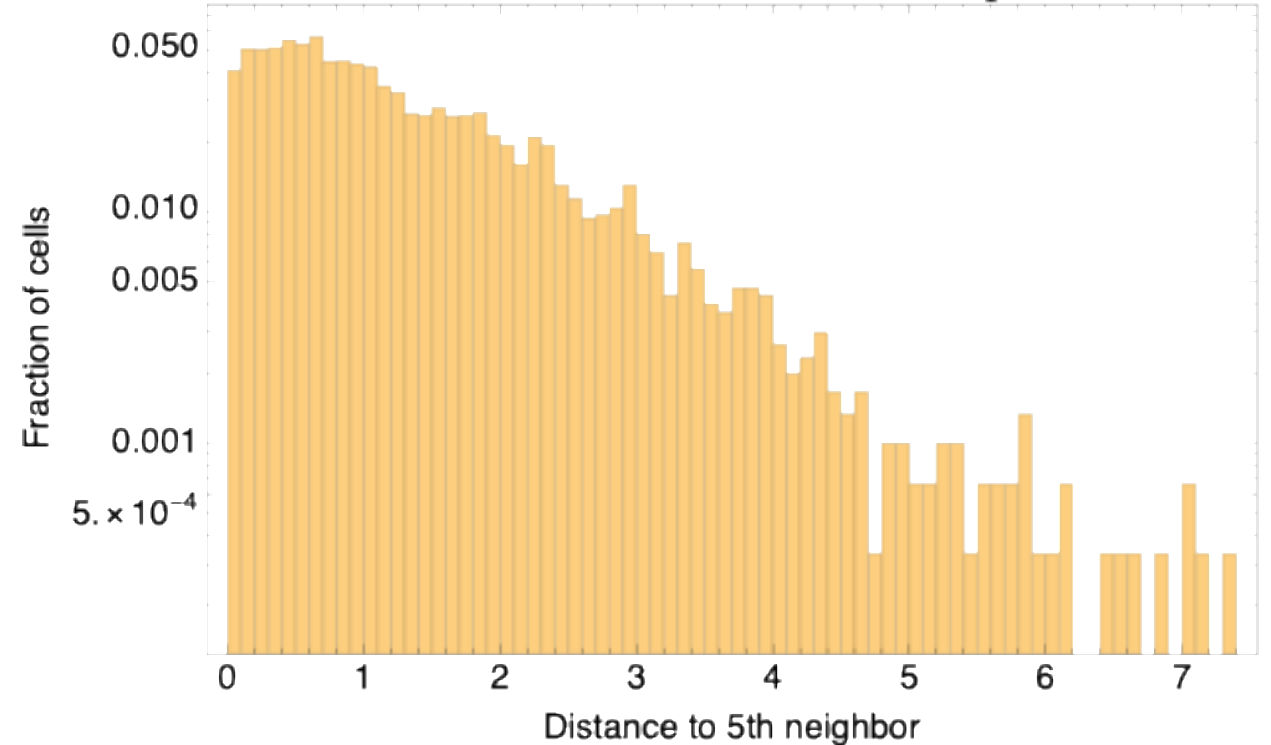
Example 3-nn graph



- For some nodes the 3rd neighbor is very nearby.
- For others even the first neighbor is very far.
- This difference is *ignored* in kNN-graphs.

In real data distances to near neighbors are widely distributed.

Distribution of distance to 5th neighbor



Cell types in the mouse cortex and hippocampus revealed by single-cell RNA-seq

AMIT ZEISEL, ANA B. MUÑOZ-MANCHADO, SIMONE CODELUPPI, PETER LÖNNERBERG, GIOELE LA MANNÒ, ANNA JURÉUS, SUELI MARQUES, HERMANY MUNGUÁ, LIQUN HE, CHRISTER BETSHOLTZ, CHARLOTTE ROLNY, GONÇALO CASTELO-BRANCO, JENS HJERLING-LEFFLER, AND STEN LINNARSSON [fewer](#) [Authors Info & Affiliations](#)

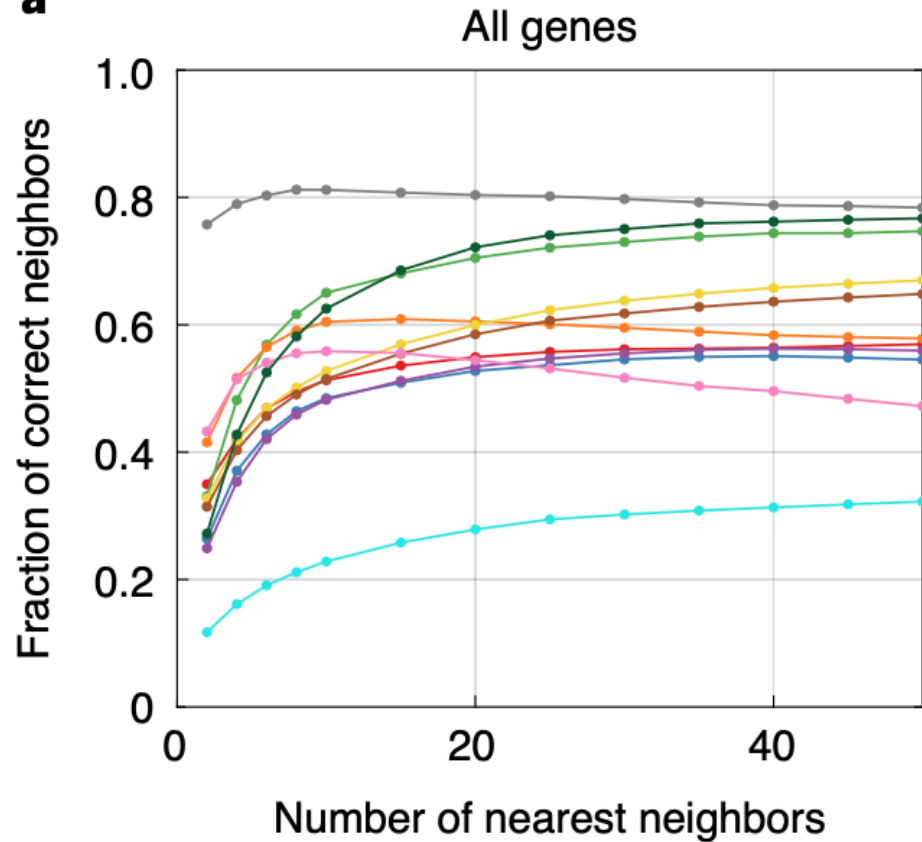
SCIENCE • 19 Feb 2015 • Vol 347, Issue 6226 • pp. 1138-1142 • DOI:10.1126/science.1234567

Case study: Why is scRNA-seq analysis hard?

1. Noisy: The number of captured mRNAs is affected by both measurement noise and intrinsic biological noise
 1. The noise depends on the absolute expression level
 2. Expression levels vary over 4-5 orders of magnitude, with most genes having low absolute expression.
 3. Variation in expression level scales with mean expression level
 4. Removing noise by dimensionality reduction can severely distort the data.
2. High-dimensional: Faithfully representing the structure of high dimensional data is difficult
 1. kNN graphs remove much structure in the data.
 2. **Correctly identifying nearest neighbors is very difficult.**
3. Hard to visualize: Current visualization methods are unreliable and uninterpretable.

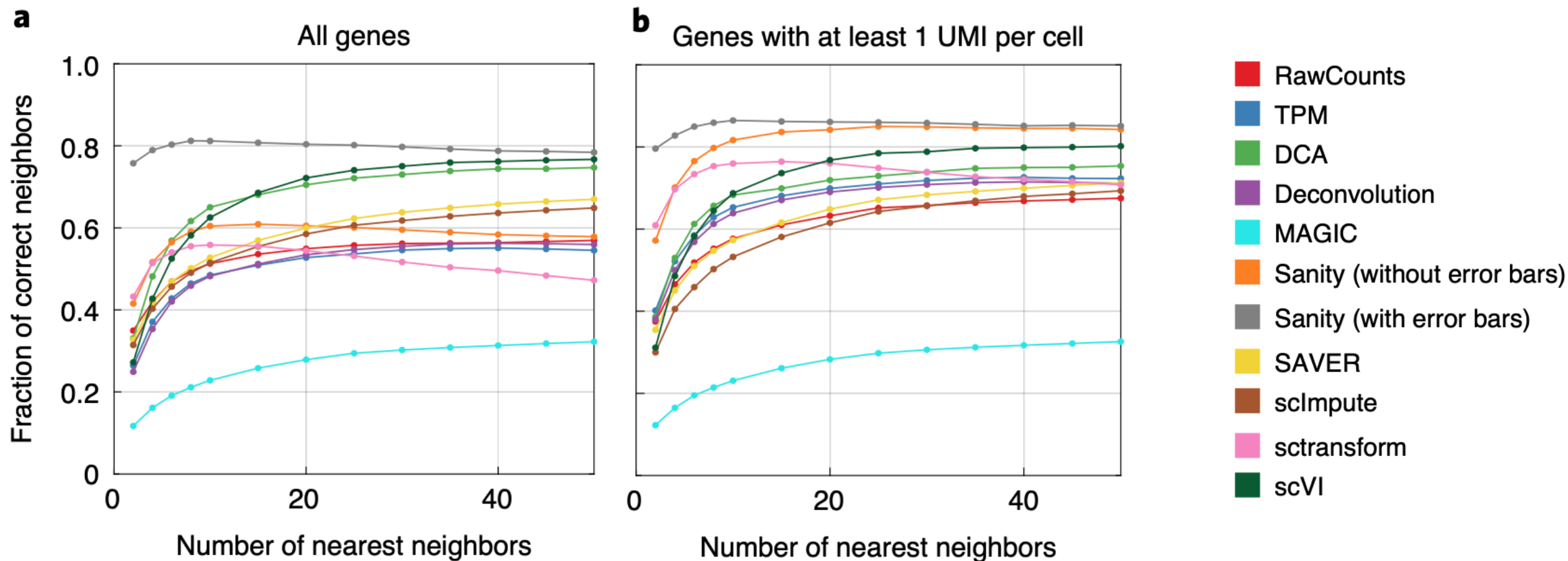
Determining the NNs is hard

a



- Simulated dataset with clear nearest neighbor structure
- *logp1* gets less than 50% of top 10 neighbors correct.
- Auto-encoders (scVI and DCA) do a bit better, but still only 70-75% correct.

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- *logp1* gets less than 50% of top 10 neighbors correct.
- Auto-encoders (scVI and DCA) do a bit better, but still only 70-75% correct.
- **Restricting to genes with higher expression helps a bit**

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Case study: How to analyze scRNA-seq data?

1. The physical observable to determine is the “log transcription quotient”

- The cell's recent history of transcription and decay rates is summarized in the *transcription activities*:

$$a_{gc} = \int_0^\infty \lambda_{gc}(t) \exp \left[- \int_0^t \mu_{gc}(\tau) d\tau \right] dt$$

- To take variations in total mRNA count N_c into account, we look at *transcription quotients*: $q_{gc} = \frac{a_{gc}}{\sum_{g'} a_{g'c}} = \left\langle \frac{n_{gc}}{N_c} \right\rangle$.
- Because variation scales with mean expression, we estimate *log transcription quotients*: $x_{gc} = \log(q_{gc}) = \log \left(\frac{a_{gc}}{\sum_{g'} a_{g'c}} \right)$

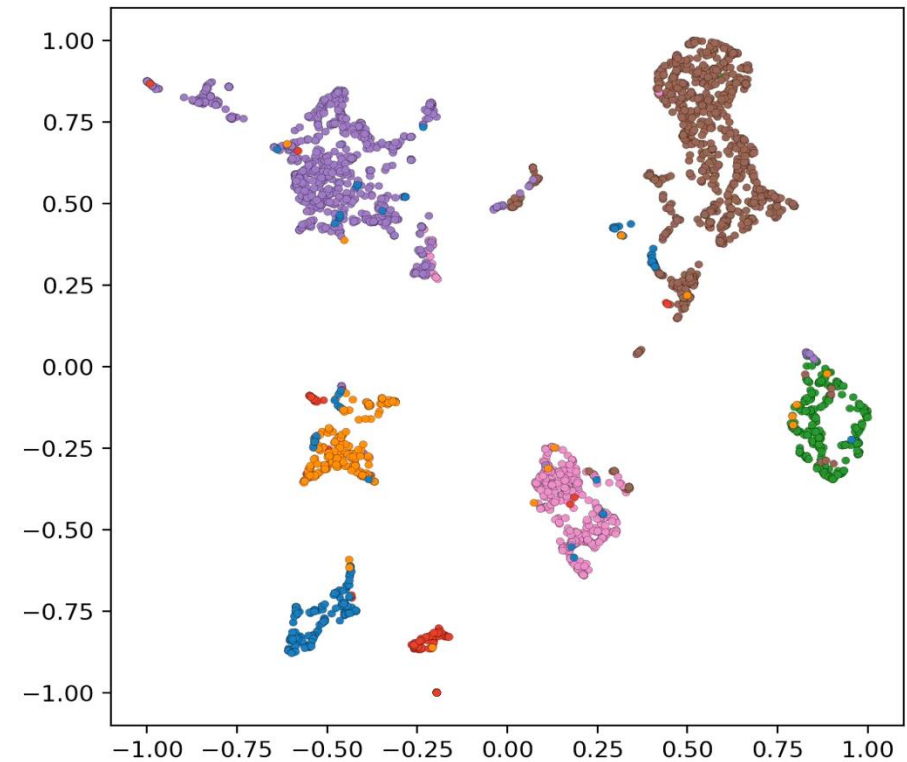
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3. The lowest count genes/cells are most noisy

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3. **Hard to visualize: Current visualization methods are unreliable and uninterpretable.**

Current visualization methods are unreliable and uninterpretable



- Stochastic (i.e. producing different answers on the same dataset).
- Uninterpretable objective function.
- Lots of parameters that can be arbitrarily tuned.
- No meaning to the axes.
- No interpretation for the shapes and positions in the resulting picture.

Case study: How to analyze scRNA-seq data?

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2. We need a better way to capture high-dimensional structure than the kNN-graph

3. The lowest count genes/cells are most noisy

4. We need a principled way of visualizing structure in scRNA-seq datasets

Case study: Our solution for scRNA-seq analysis

1. **Preprocessing:** Map raw reads to get mRNA-counts
2. **Cellstates:** Simplify the data by grouping all cells whose gene expression states are *statistically indistinguishable*. Combine the mRNA counts n_{gc} of the cells in each *cellstate*.
3. **Sanity:** Estimate the LTQs x_{gc} of each *cellstate* with rigorous *uncertainty estimates* δx_{gc} .
4. **Bonsai:** Represent the structure in the data by constructing a *tree structure* relating all *cellstates*.

But first: a 5-minute break

Today's program

- **14:00 – 14:30:** Challenges in scRNA-seq data analysis
- **14:30 – 15:00:** Our solution: *Sanity* + *Bonsai*
- **15:00 – 16:15:** Hands-on tutorial on running *Bonsai* and *Bonsai-scout*
- **16:20 – 16:50:** From dataset to paper figure
- **17:00 – 18:00:** Q&A + individual dataset analysis

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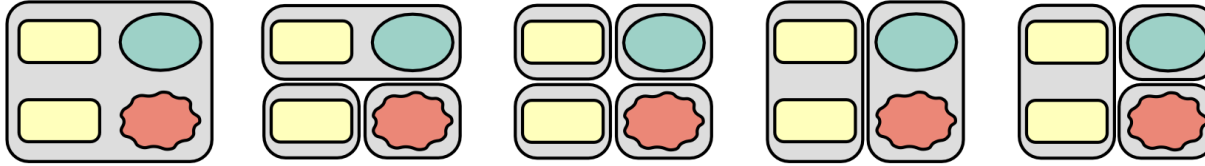
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Cellstates: grouping cells with statistically indistinguishable gene expression states

PARTITIONS ρ OF CELLS INTO CELLSTATES

Sorted by posterior likelihood $P(D|\rho)$



Low likelihood
partition

High likelihood
partition

Identifying cell states in single-cell RNA-seq data at statistically maximal resolution

Pascal Grobecker,  Erik van Nimwegen

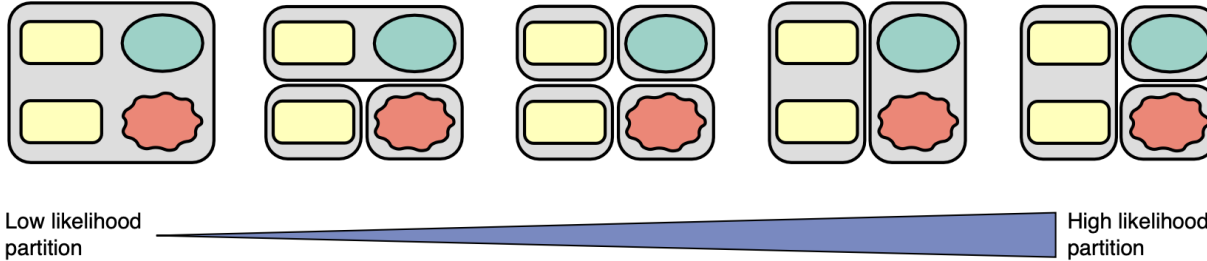
doi: <https://doi.org/10.1101/2023.10.31.564980>

- **We exploit that we know the statistics of biological + measurement noise.**
- For a group of cells s , the probability of their total UMI counts $\vec{n}_s$ given their gene expression state vector of transcription quotients $\vec{q}$ is multinomial: $P(\vec{n}_s|\vec{q}) = N_s! \prod_g \frac{1}{n_{gs}!} (q_g)^{n_{gs}}$

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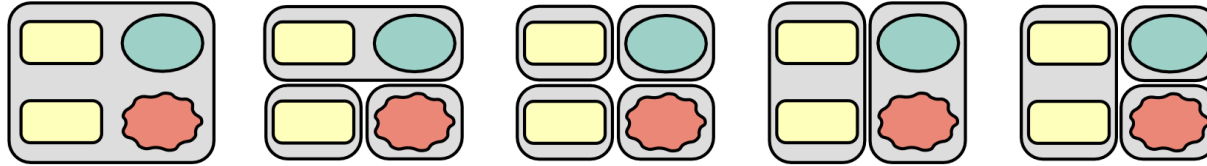
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- We marginalize over the unknown transcription quotient vector $\vec{q}$ to obtain: $P(\vec{n}_s|\vec{\theta}) = \frac{\Gamma(\Theta)}{\Gamma(N_s+\Theta)} \prod_g \frac{\Gamma(n_{gs}+\theta_g)}{\Gamma(\theta_g)}$, with θ_g parameters of a prior distribution over transcription quotients, which is set by the sum of UMI counts over all cells.

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- In this way, a likelihood $P(D|\rho)$ of the data is determined for any partition ρ of the cells into subsets.
- The Cellstates algorithm uses Markov Chain Monte Carlo to find the optimal partition ρ .

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A popular option: **logp1-normalization**

How to estimate these “LTQs”
from the mRNA-counts?

- Note that: $\langle n_{gc} \rangle = N_c e^{x_{gc}}$
- This suggests we may estimate x_{gc} just as: $x_{gc} = \log\left(\frac{n_{gc}}{N_c}\right)$ but this is not defined when $n_{gc} = 0$.
- Ad hoc fix: Add an offset and rescale by the average (or median) total UMI count $N = \langle N_c \rangle$: $x'_{gc} = \log\left(N \frac{n_{gc}}{N_c} + 1\right)$
- But the actual count n_{gc} differs from the expectation $\langle n_{gc} \rangle$ by Poisson noise!

Sanity: estimating LTQs accounting for Poisson sampling noise

SANITY is available at: <https://github.com/jmbreda/Sanity>

Input:

- Any matrix with mRNA counts n_{gc} across genes g and cells c .

Output:

- The average log-transcription-quotient μ_g for each gene g .
- The estimated variance in log transcription-quotient across cells: v_g
- The log fold-change δ_{gc} for each gene in each cell.
- An uncertainty estimate, ε_{gc} for each log fold-change.

Bayesian inference of gene expression states from single-cell RNA-seq data

[Jérémie Breda](#), [Mihaela Zavolan](#) & [Erik van Nimwegen](#) 

[Nature Biotechnology](#) **39**, 1008–1016 (2021) | [Cite this article](#)

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- An error bar ε_{gc} for each log fold-change.

Note:

No tunable parameters. Only assumption is the Gaussian (i.e. max-ent) prior on log-fold changes.

Bayesian inference of gene expression states from single-cell RNA-seq data

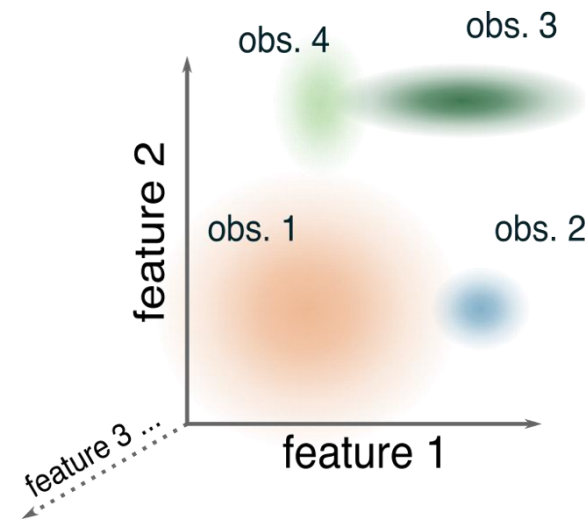
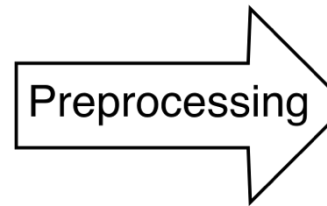
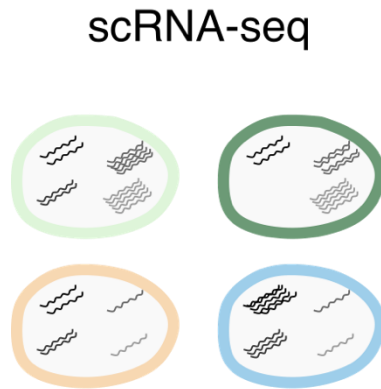
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Case study: Our solution for scRNA-seq analysis

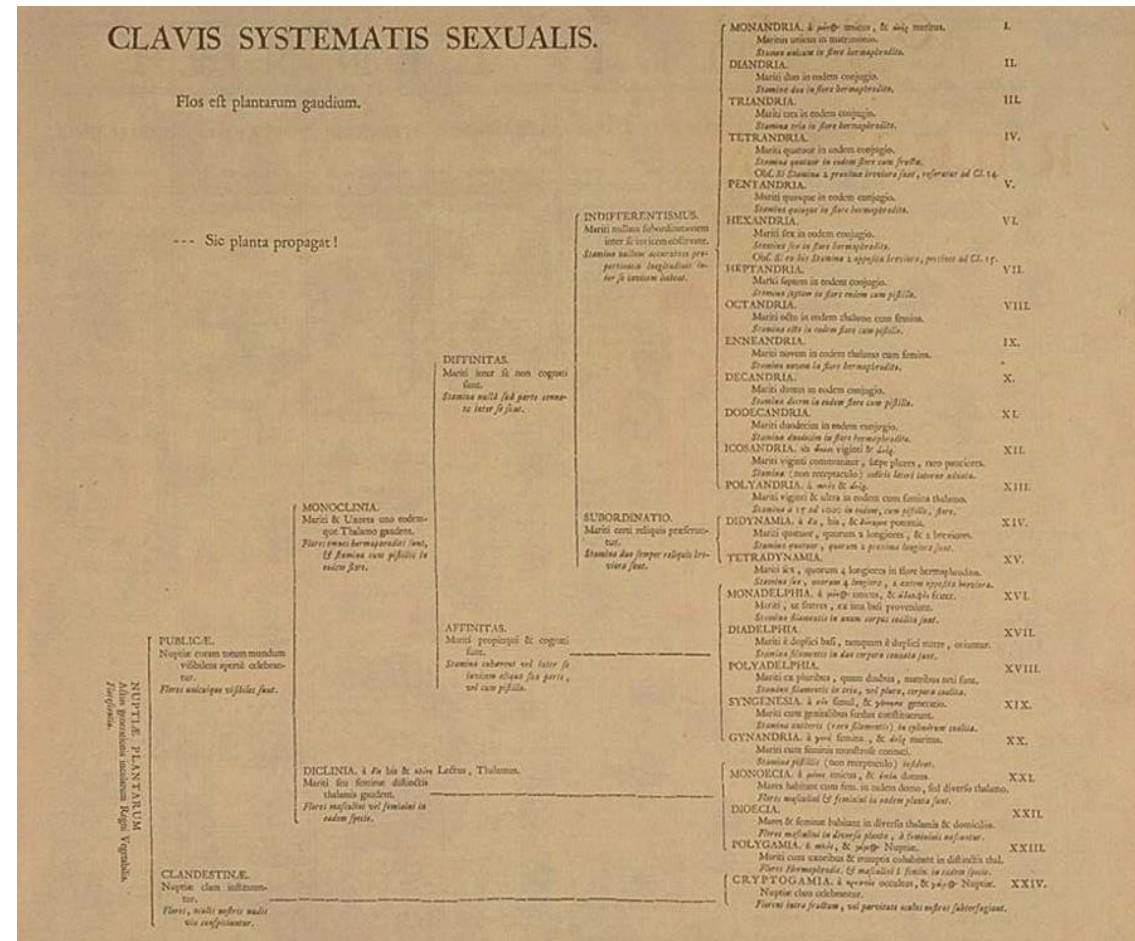
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Bonsai: How to represent high-dimensional structure, even if the data are properly normalized?



How to explore >20,000 dimensions?

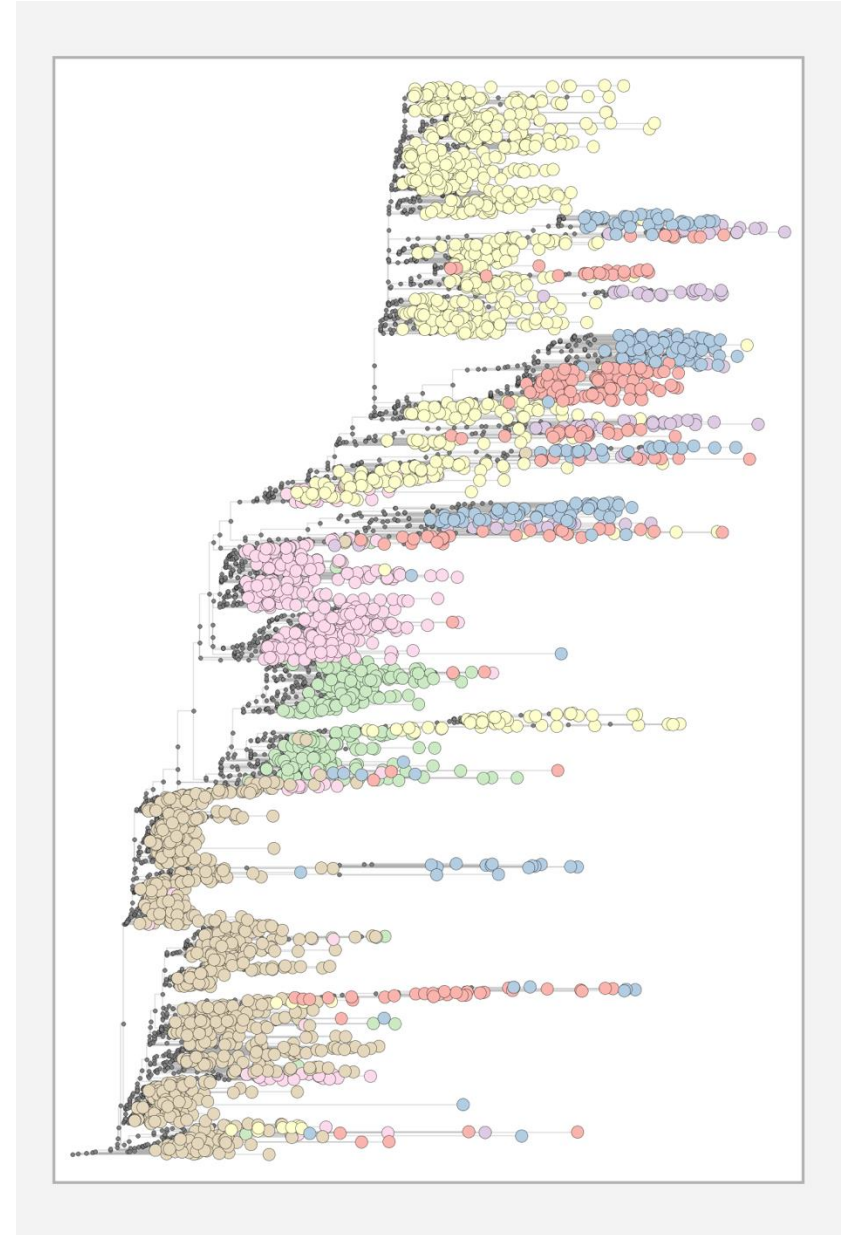
Trees can help!



Linnaeus' tree of the sexual system of plants, 1735

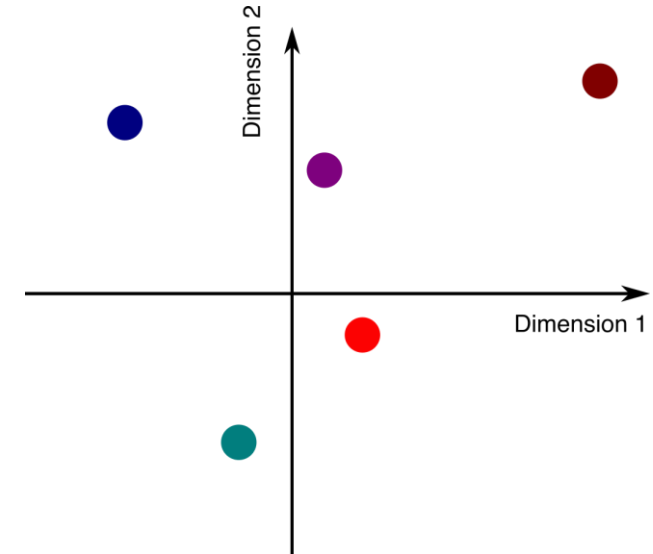
How to explore $>20,000$ dimensions?
Trees can help!

Reason 1: The blessing of dimensionality

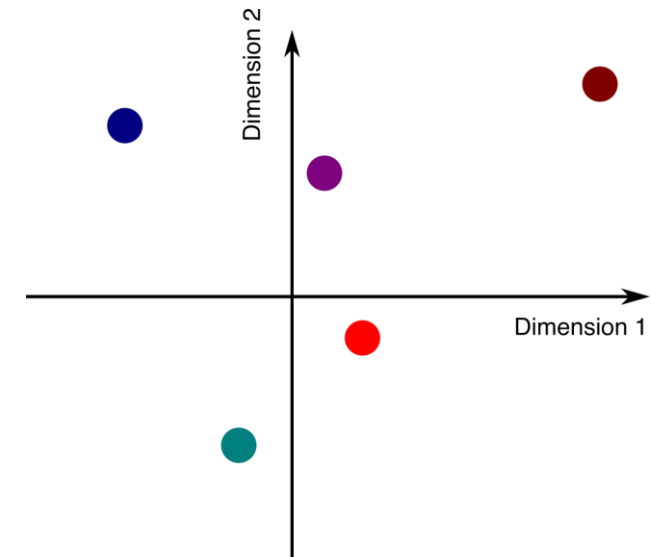
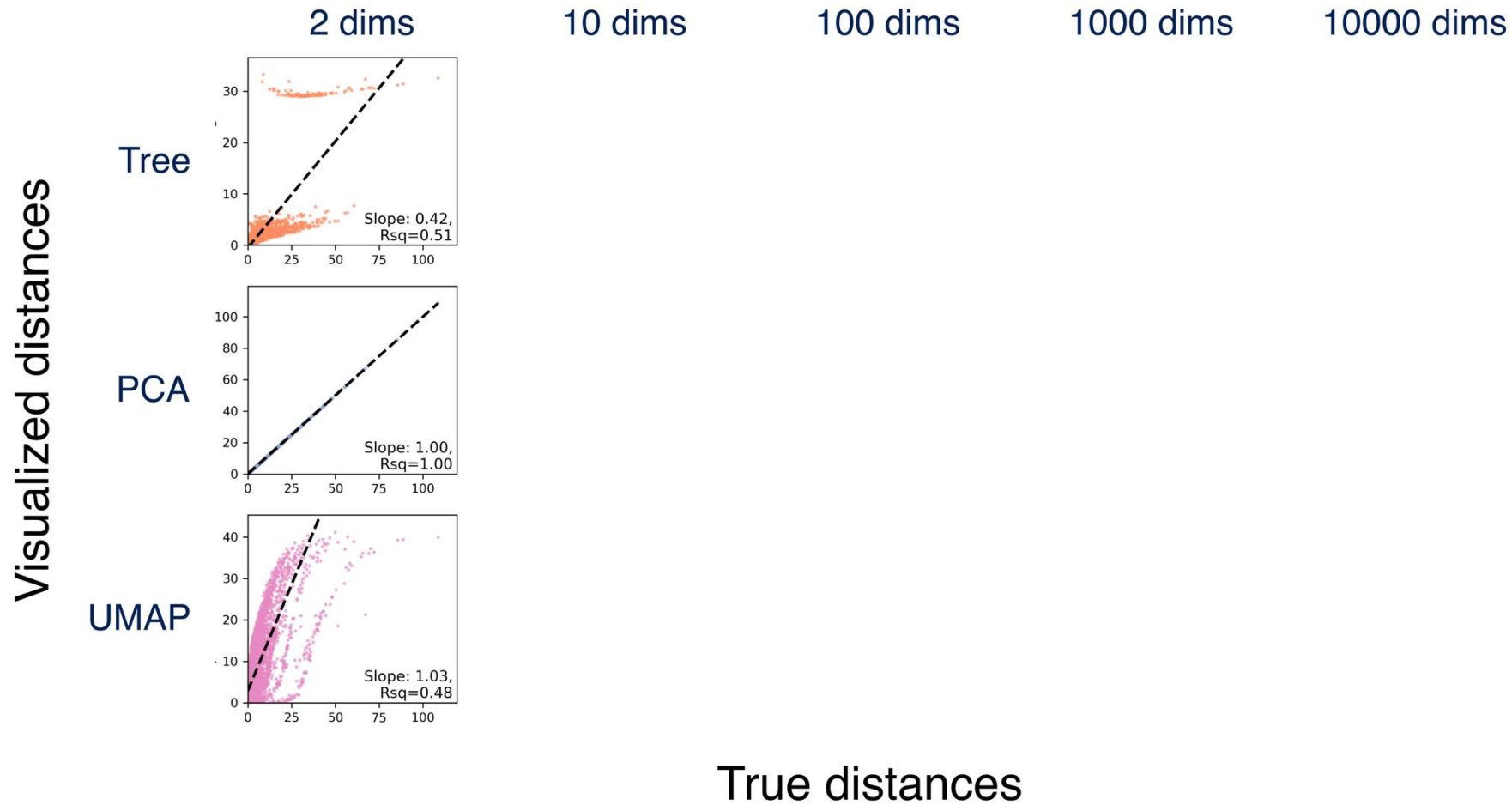


Trees benefit from the “blessing of dimensionality”

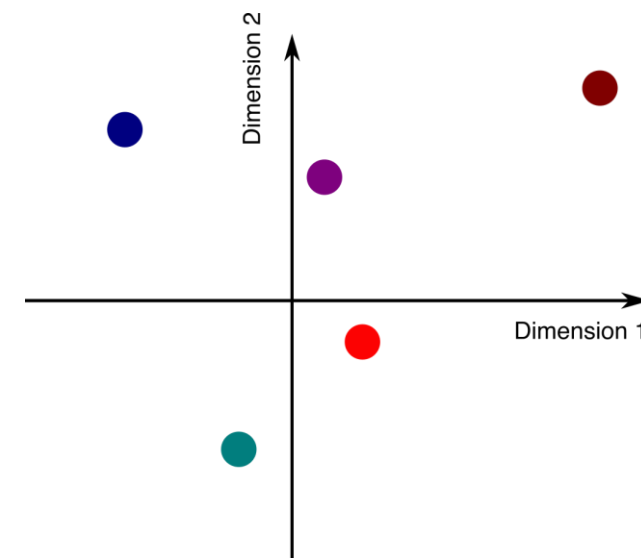
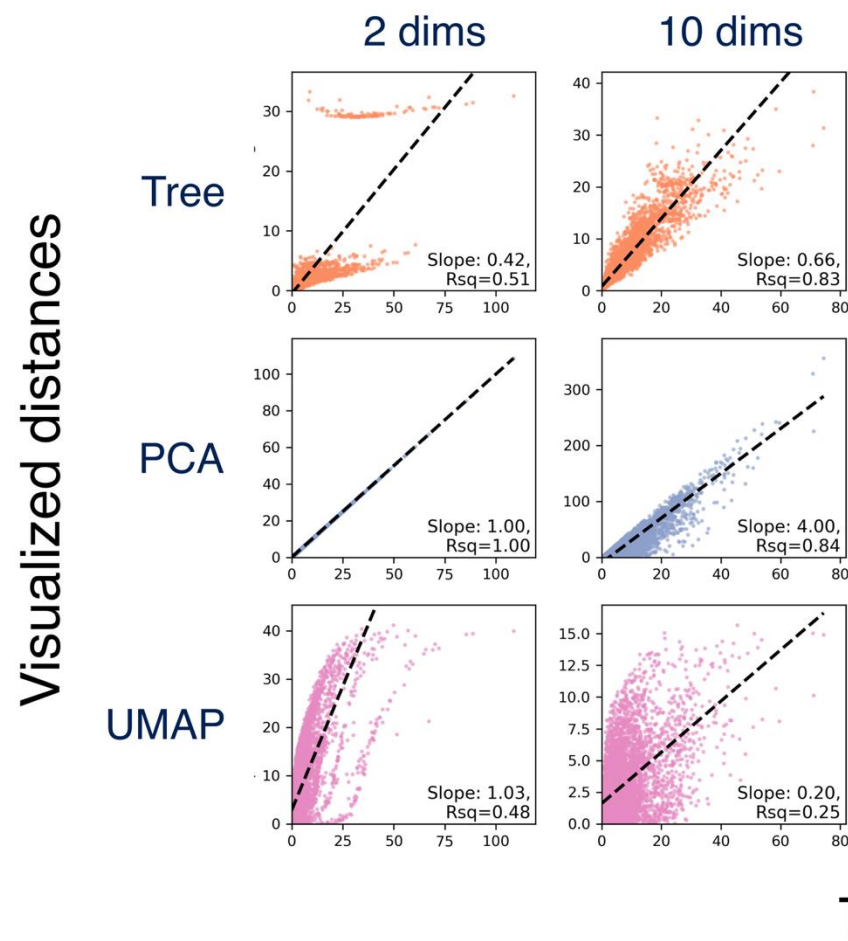
1. Take 100 points from a spherical Gaussian
 - in 2, 10, 100, 1000, 10000 dimensions



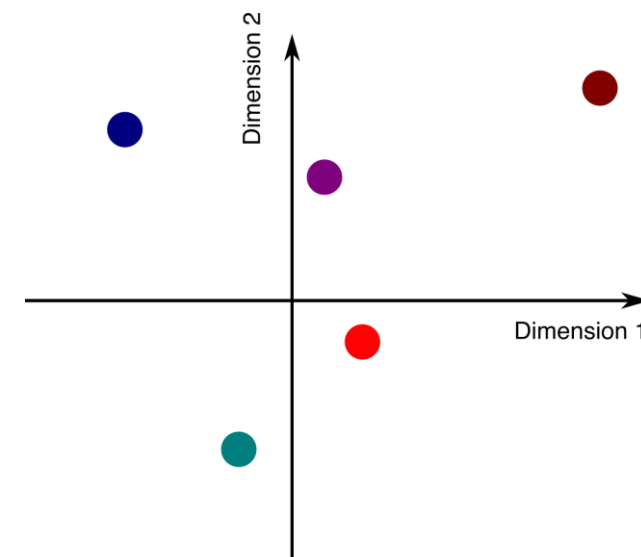
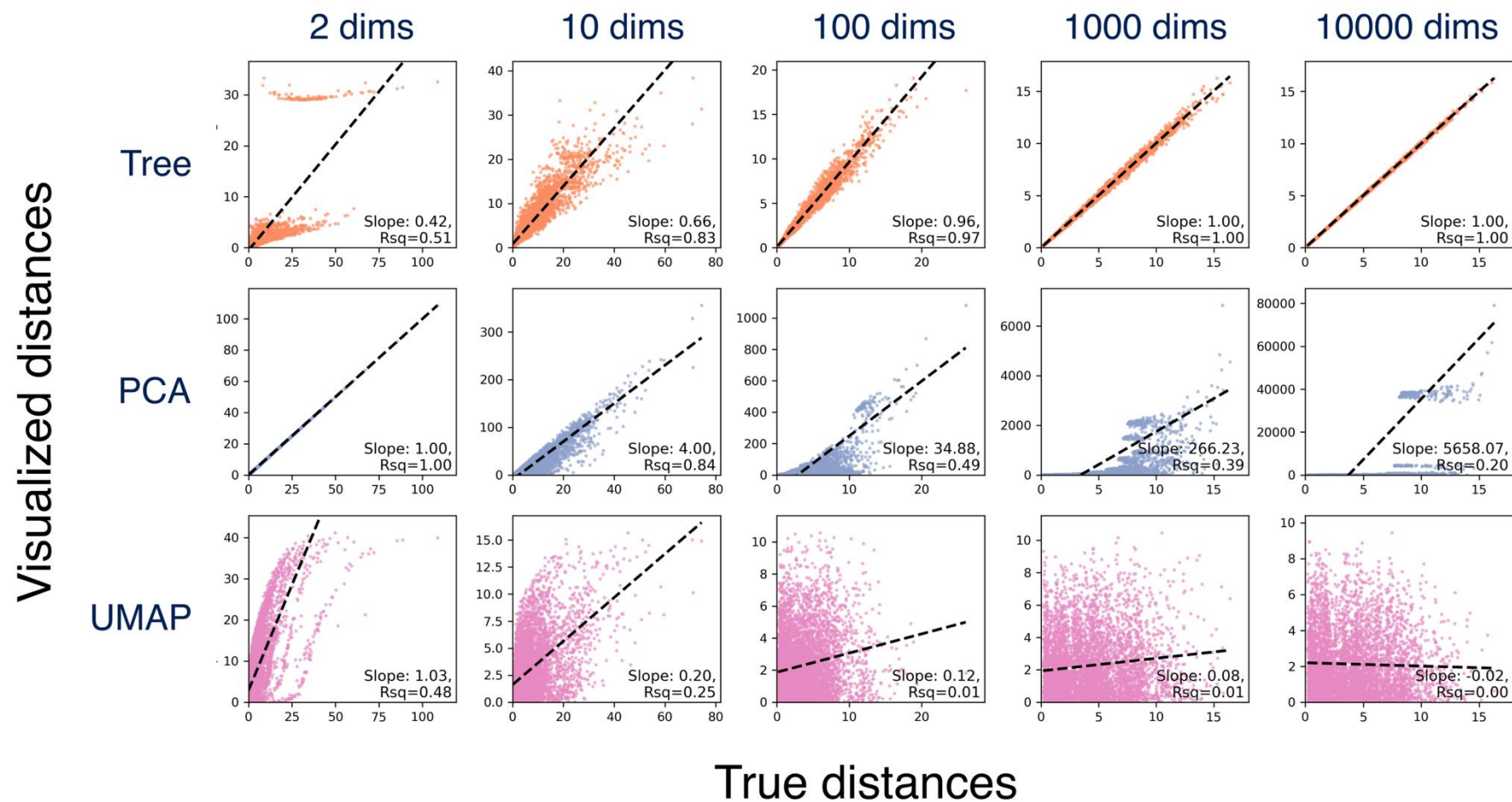
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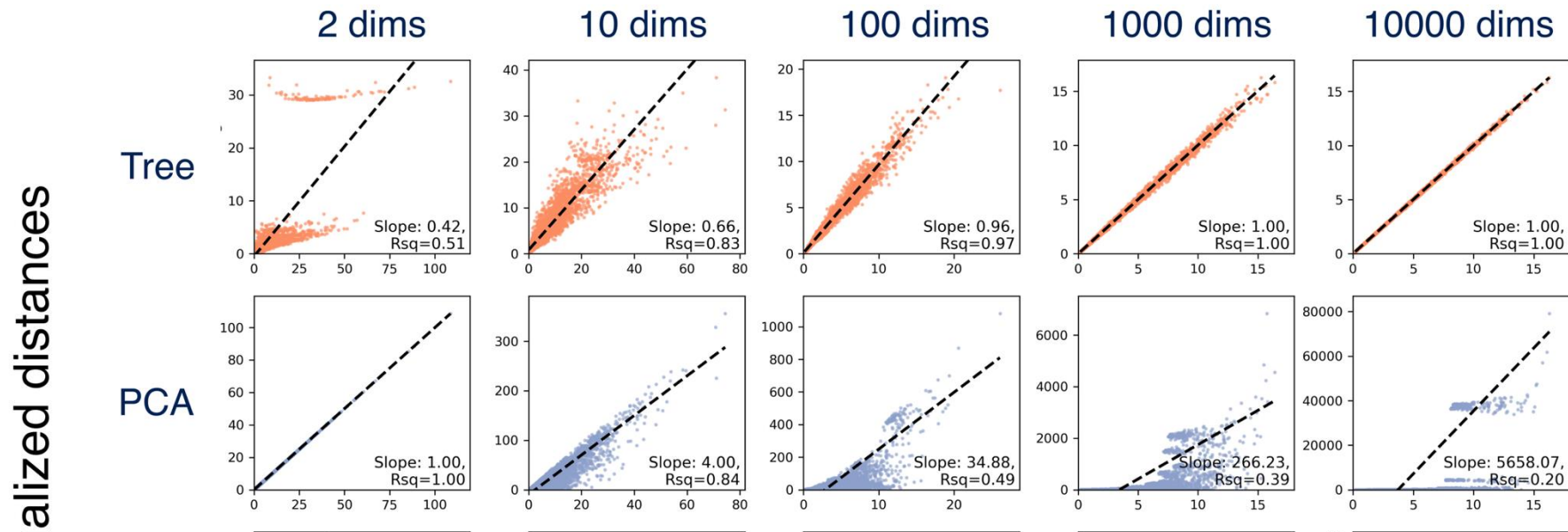
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Blessing of dimensionality

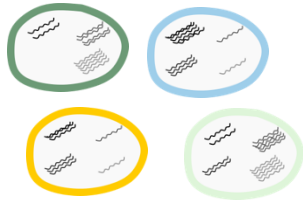
In high-dimensional spaces $O(n^2)$ pairwise distances can **generically** be captured in $O(n)$ branch lengths

How to explore >20,000 dimensions?

Trees can help!

Reason 1: The blessing of dimensionality

Reason 2: Cells are naturally related by a lineage tree

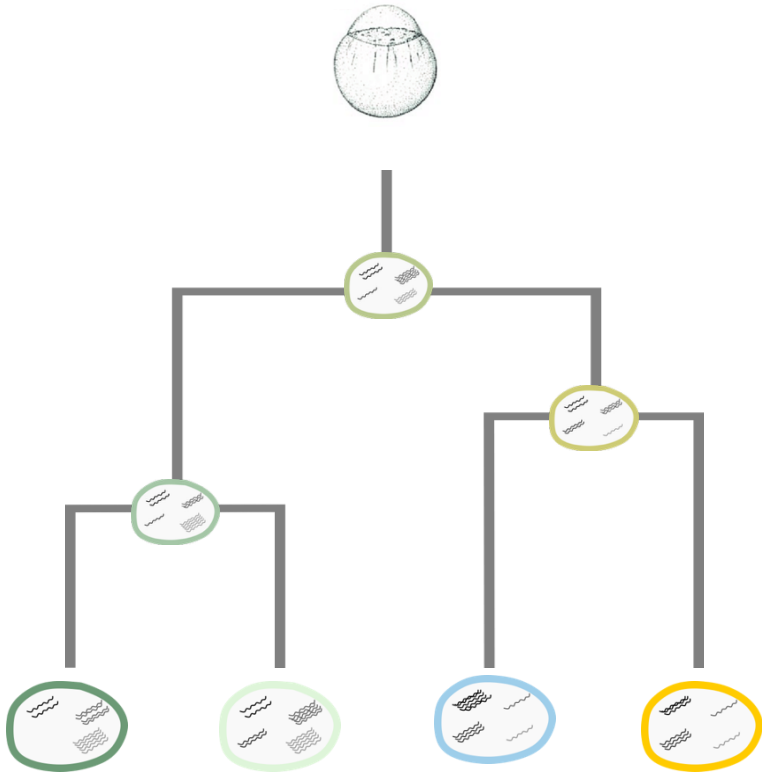


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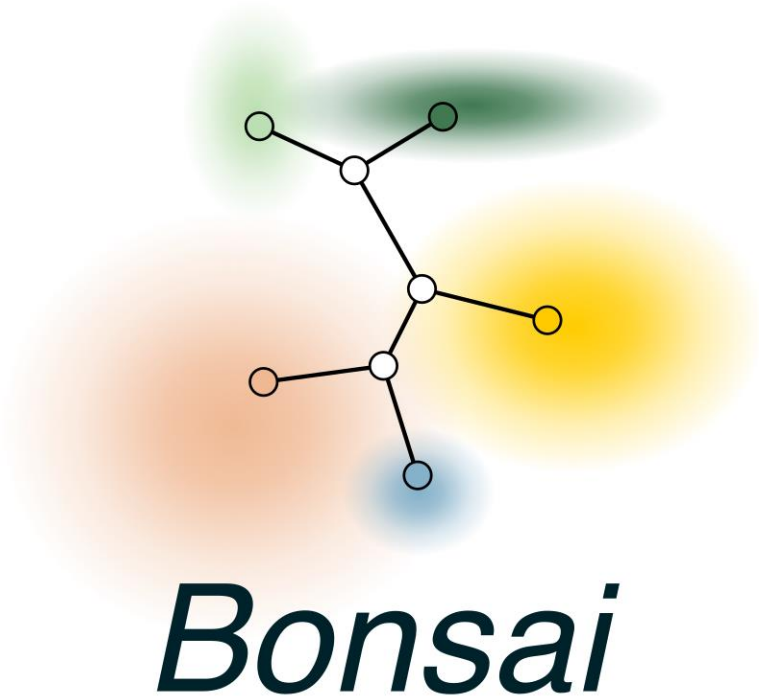


Bonsai infers the maximum likelihood single-cell tree



Bonsai

Bonsai infers the maximum likelihood single-cell tree



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***Bonsai*: Tree representations for distortion-free visualization and exploratory analysis of single-cell omics data**

 Daan H. de Groot,  Sarah X. Morillo Leonardo,  Mikhail Pachkov,  Erik van Nimwegen

doi: <https://doi.org/10.1101/2025.05.08.652944>

This article is a preprint and has not been certified by peer review [what does this mean?].

Abstract

Full Text

Info/History

Metrics

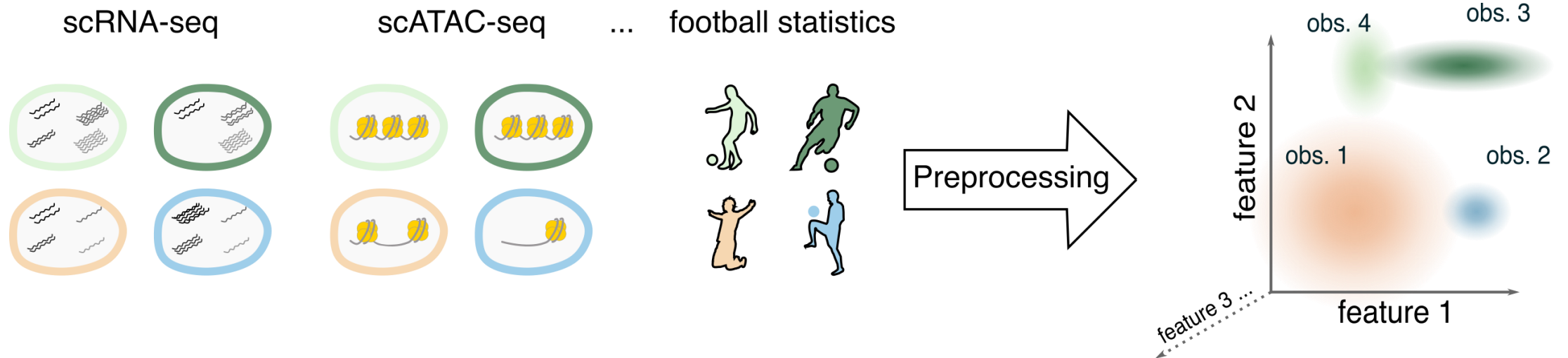
 [Preview PDF](#)

Currently under revision

Bonsai infers the maximum likelihood single-cell tree

The *Bonsai*-ingredients:

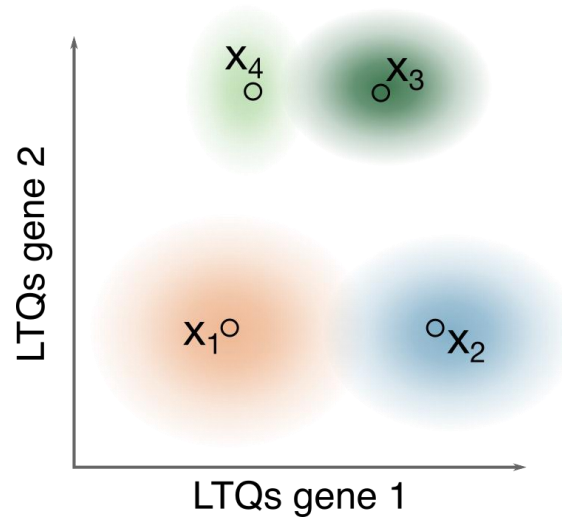
1. Preprocessing raw data to infer normalized features and uncertainty estimates



Bonsai infers the maximum likelihood single-cell tree

The *Bonsai*-ingredients:

1. Preprocessing raw data to infer normalized features and uncertainty estimates
2. **Defining the likelihood of the data given any tree**

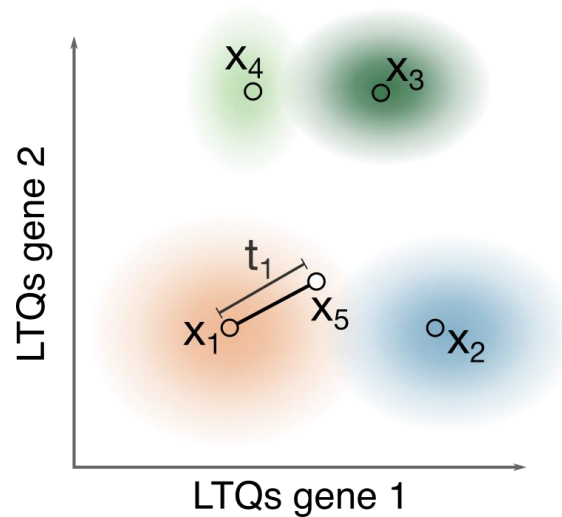


$$\prod_{i \in \text{cells}} P(\text{cell}_i \mid x_i)$$

Bonsai infers the maximum likelihood single-cell tree

The *Bonsai*-ingredients:

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We take a conservative diffusion prior for movement in gene expression space.

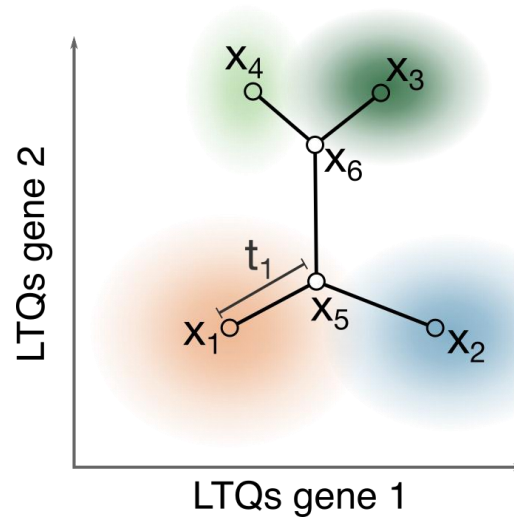
The diffusion time is proportional to the tree-branch length

$$\prod_{i \in \text{cells}} P(\text{cell}_i \mid x_i) \times \prod_{j \in \text{nodes}} P(x_j \mid x_{\pi(j)}, t_j)$$

Bonsai infers the maximum likelihood single-cell tree

The *Bonsai*-ingredients:

1. Preprocessing raw data to infer normalized features and uncertainty estimates
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$$P(T) = \int \cdots \int \left(\prod_{i \in \text{cells}} P(\text{cell}_i \mid x_i) \times \prod_{j \in \text{nodes}} P(x_j \mid x_{\pi(j)}, t_j) \right) dx_1 \cdots dx_N$$

Bonsai infers the maximum likelihood single-cell tree

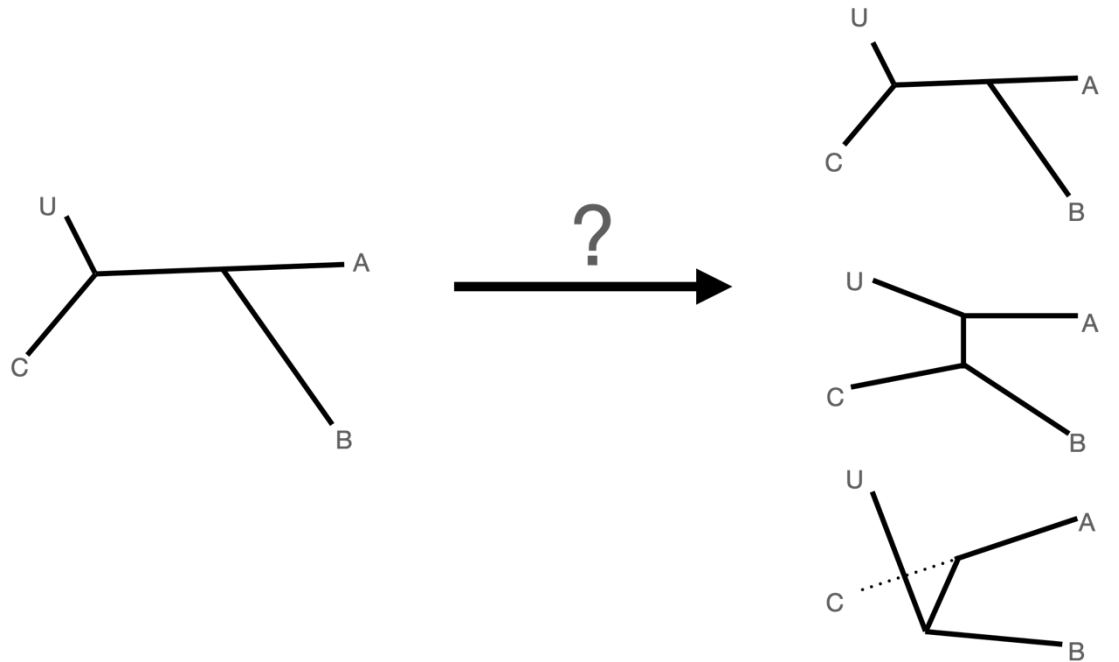
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3. **Finding the maximum likelihood tree**

Bonsai infers the maximum likelihood single-cell tree

The *Bonsai*-ingredients:

1. Preprocessing raw data to infer normalized features and uncertainty estimates
2. Defining the likelihood of the data given any tree
3. Finding the maximum likelihood tree
 1. Start from a star-tree
 2. Optimize the branch lengths
 3. Merge pairs greedily
 4. Interchange nearest-neighbors

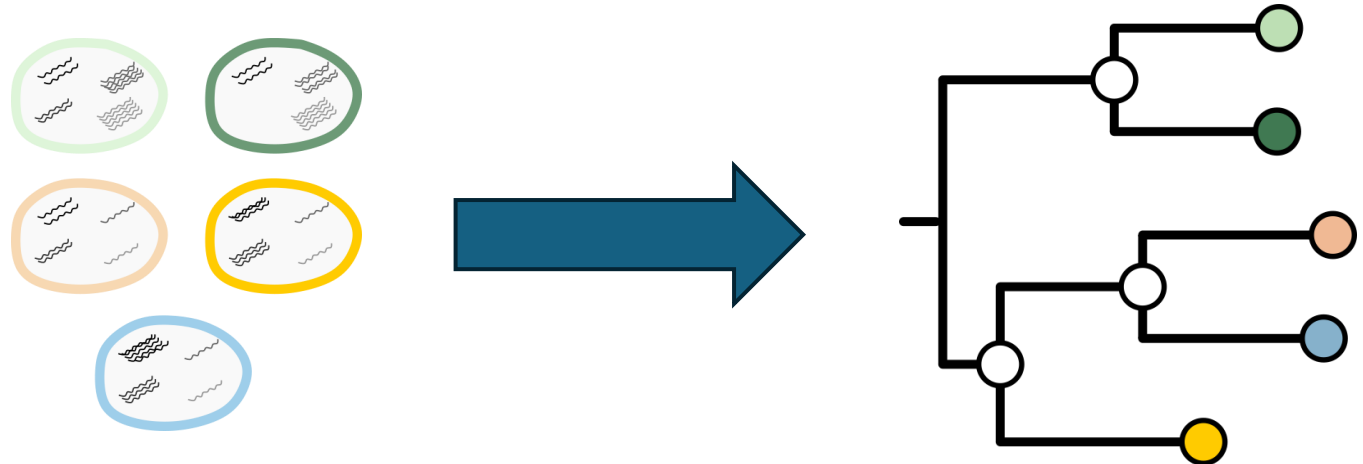


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4. Visualize the tree

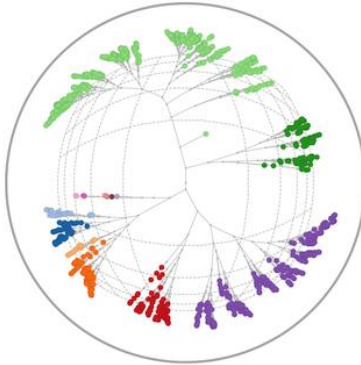


But does *Bonsai* work?

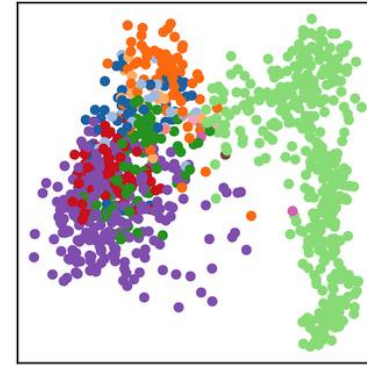
- ☐ Does it work on simulated data?
- ☐ Does it scale to real-sized datasets?
- ☐ Is it user-friendly?
- ☐ Does it produce novel biological results?

Bonsai outperforms any visualization method on simulated data

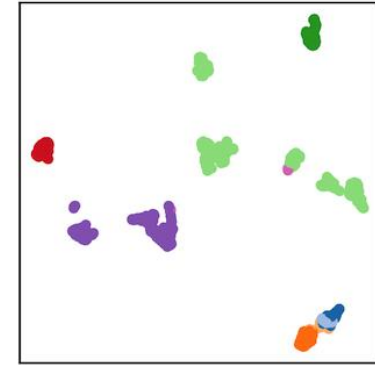
Ground truth



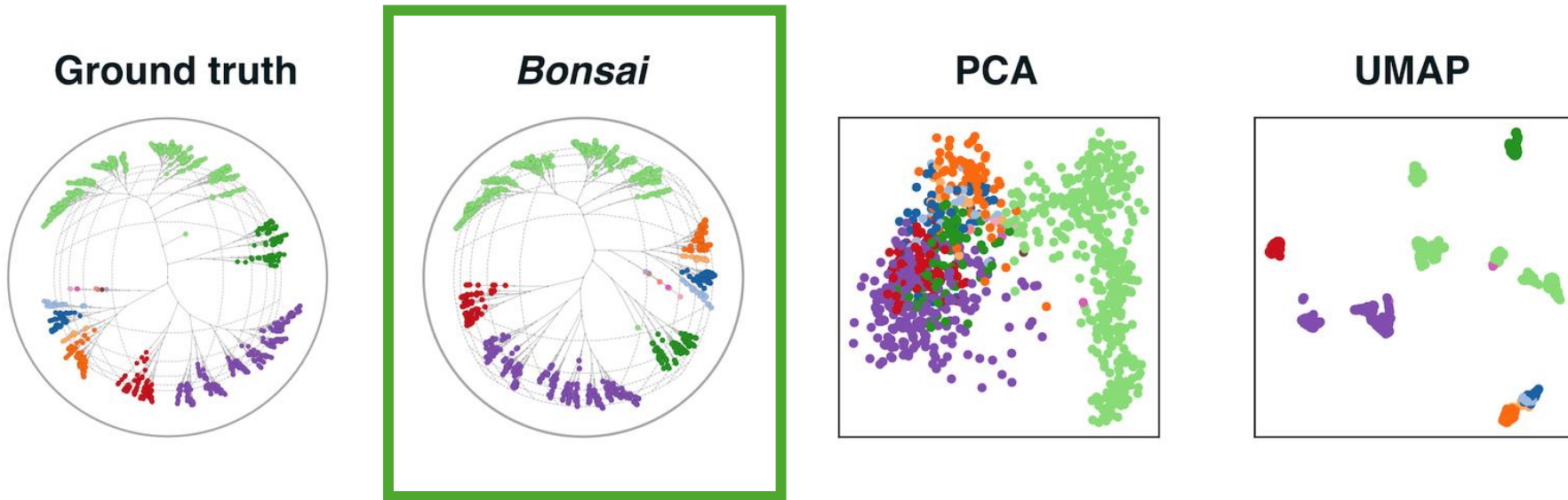
PCA



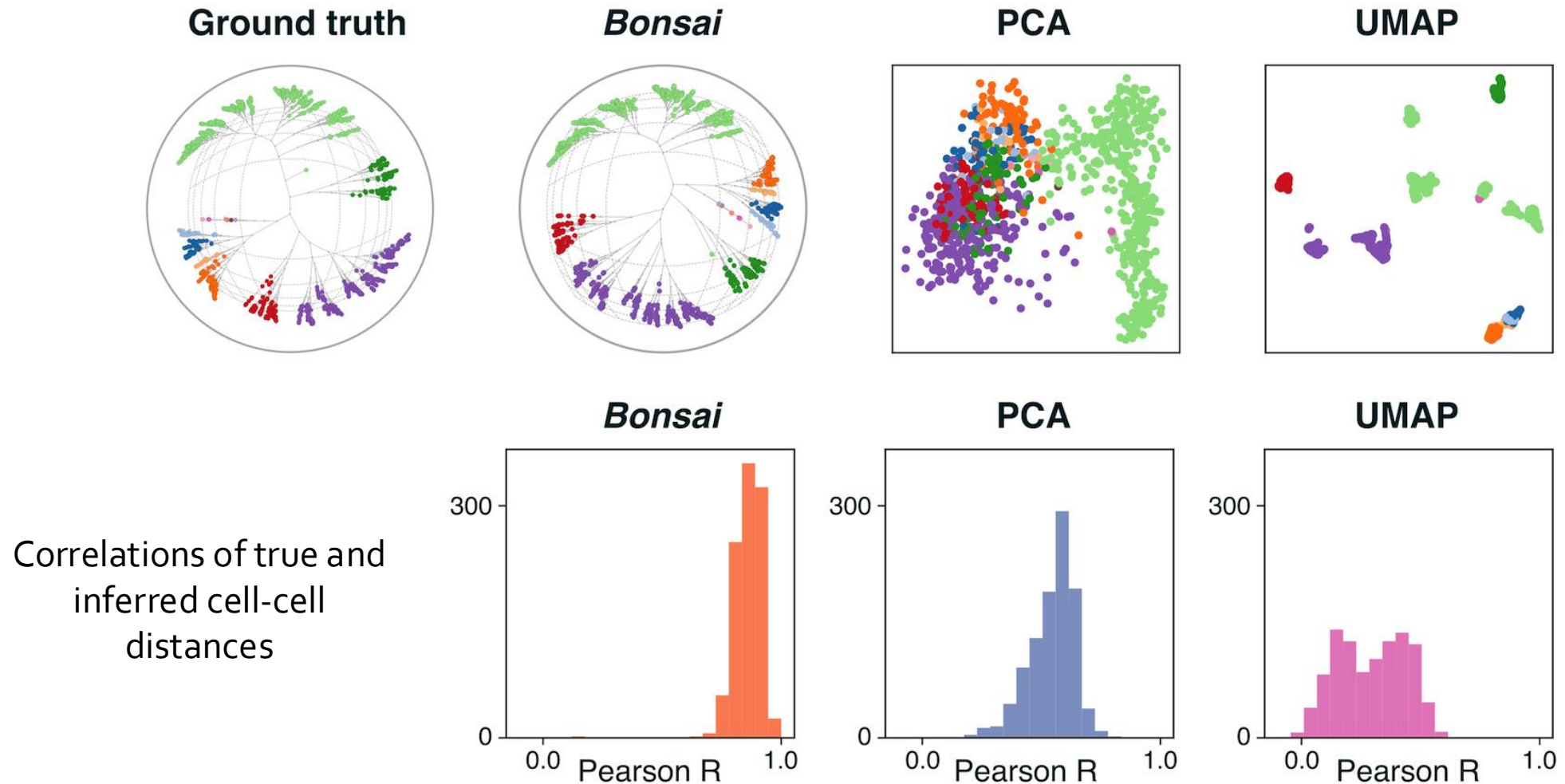
UMAP



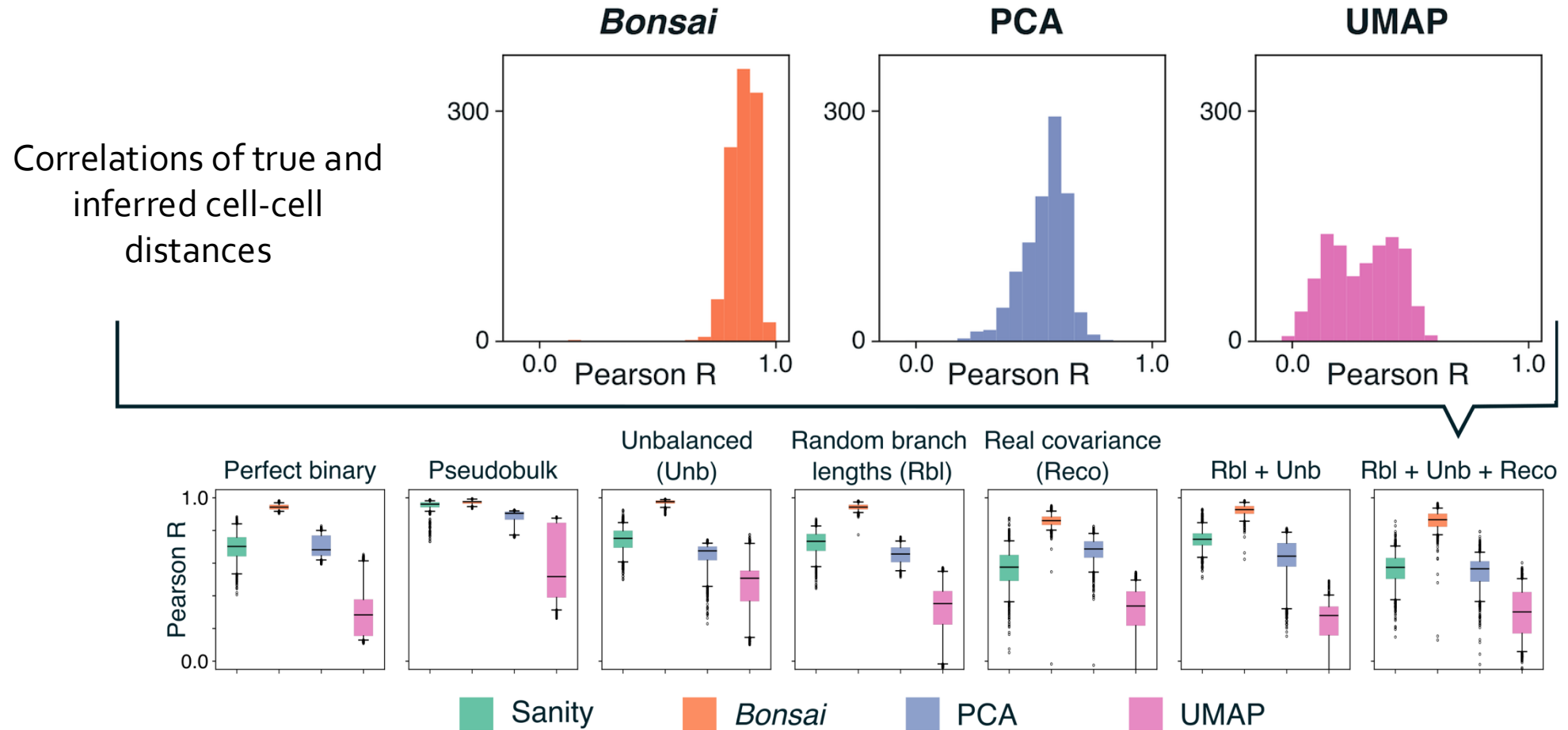
Bonsai outperforms any visualization method on simulated data



Bonsai visualizes accurate pairwise distances on all length-scales



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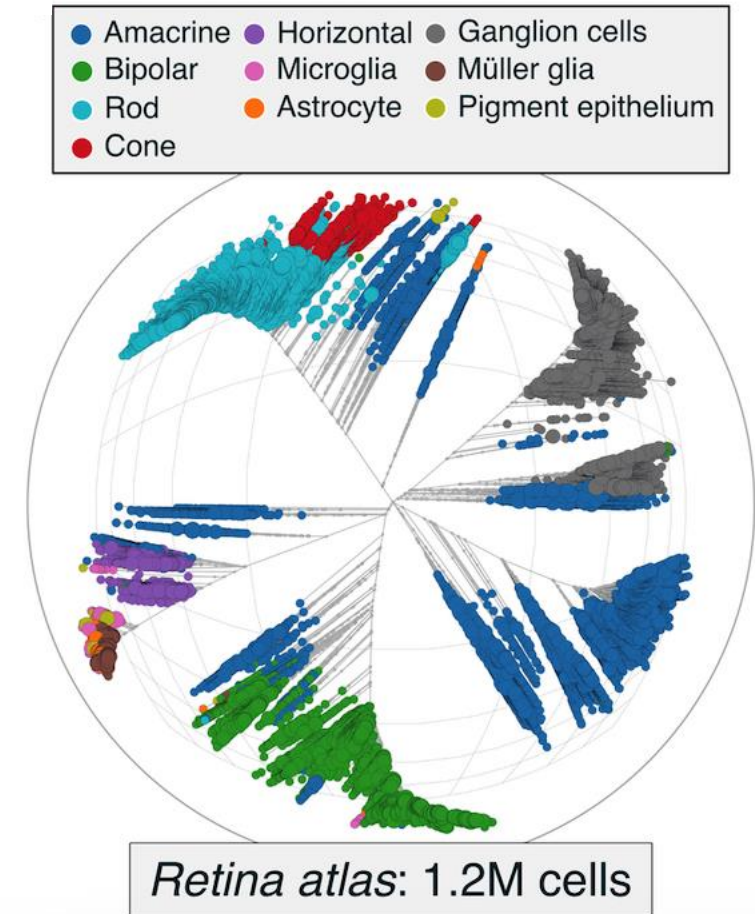


But does *Bonsai* work?

- ✓ Does it work on simulated data?
- ☐ Does it scale to real-sized datasets?
- ☐ Is it user-friendly?
- ☐ Does it produce novel biological results?

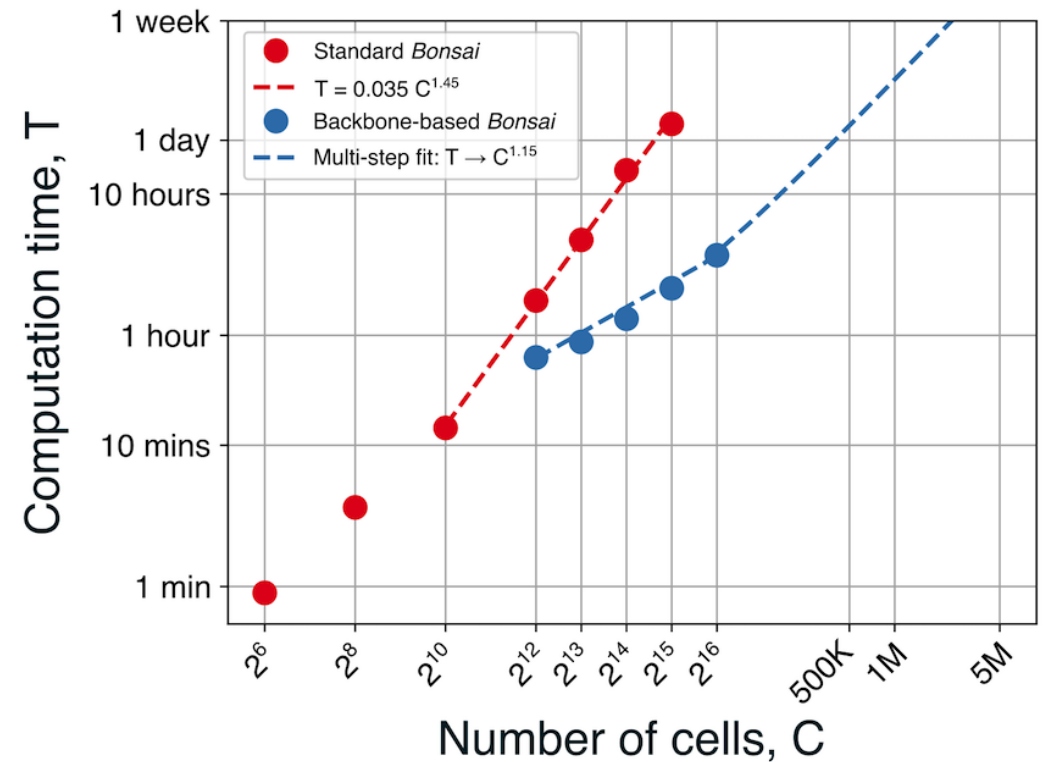
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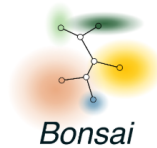
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We created a server for the analysis of scRNA-seq data

Try it out at bonsai.unibas.ch



Bonsai analysis of scRNA-seq data



Data upload.

Submit TSV mRNA count table

Submit MTX mRNA count table

Email address (optional)

Email

...

?

Project name (optional)

Project name

?

Select mRNA count table (.tsv)

Choose File

No file chosen

?

Select annotation file(s)

Choose File

No file chosen

?

Submit



This portal is operated by the van Nimwegen group of the Biozentrum,
University of Basel and Swiss Institute of Bioinformatics.



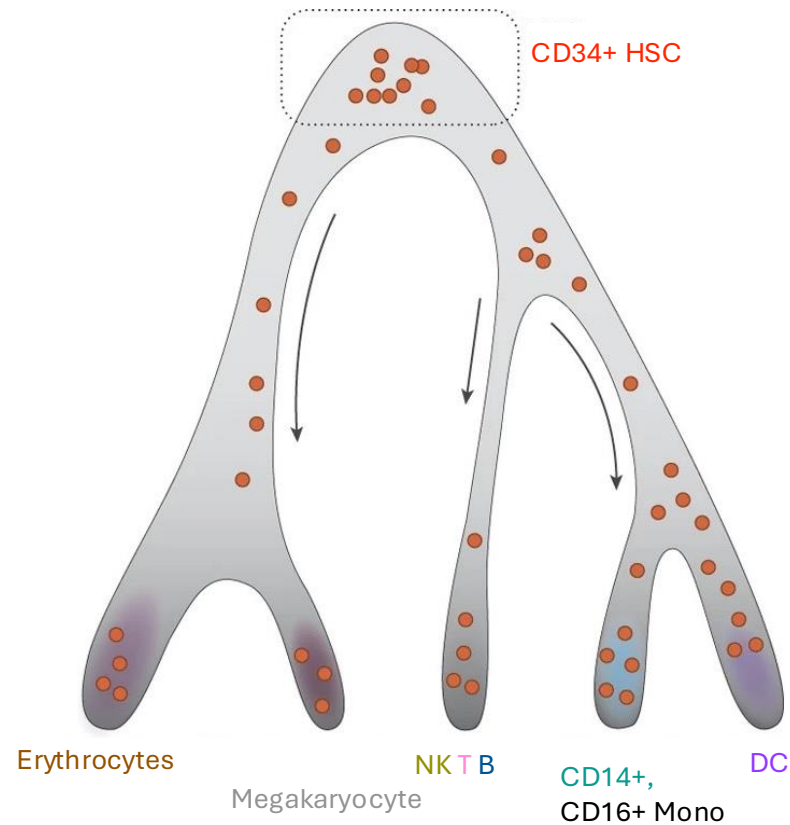
And we make results available in *Bonsai-scout*



But does *Bonsai* work?

- ✓ Does it work on simulated data?
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Bonsai reconstructs correct differentiation trajectories on a blood-cell dataset **without tuning any parameter**

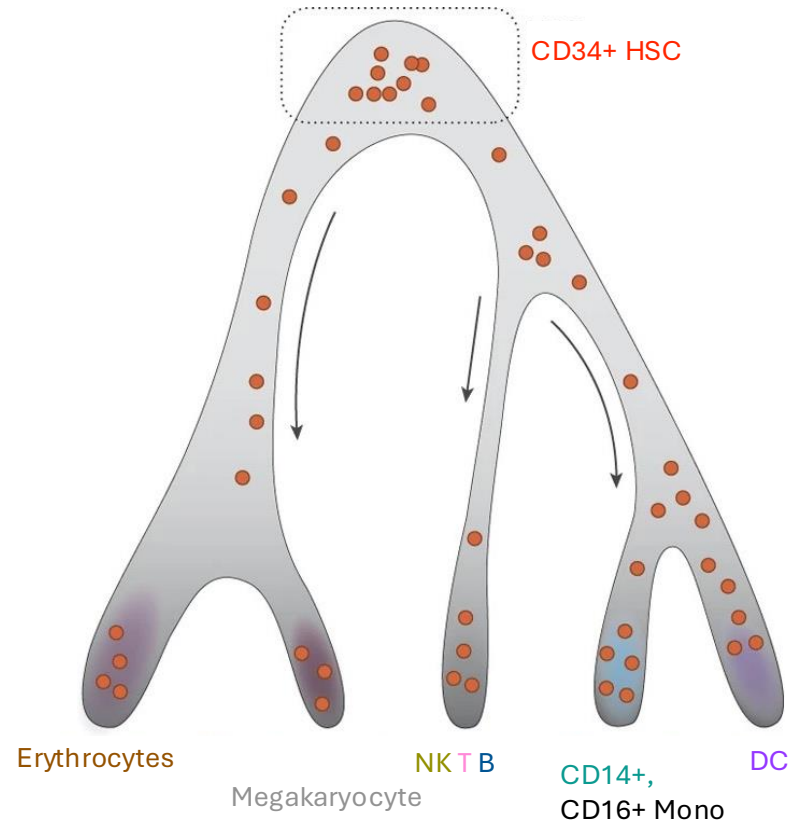


Sarah Morillo

Laurent and Göttgens, *Nature*, 2018

- 7,509 cord blood mononuclear cells
(Stoeckius et al. 2017)

Bonsai reconstructs correct differentiation trajectories on a blood-cell dataset **without tuning any parameter**



Laurent and Göttgens, *Nature*, 2018

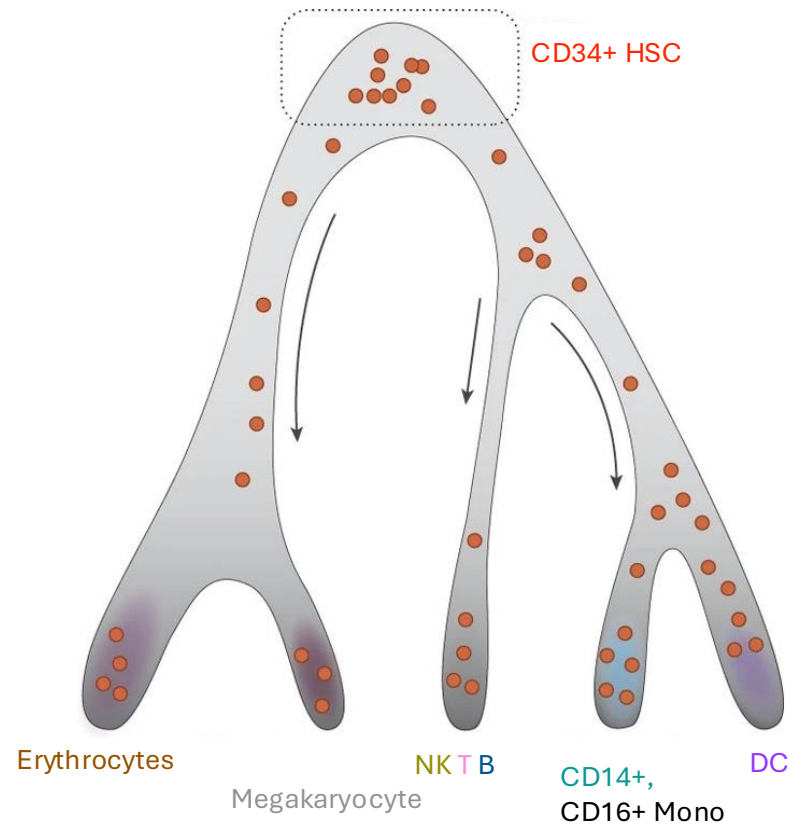


- 7,150 cord blood mononuclear cells
(Stoeckius et al. 2017)

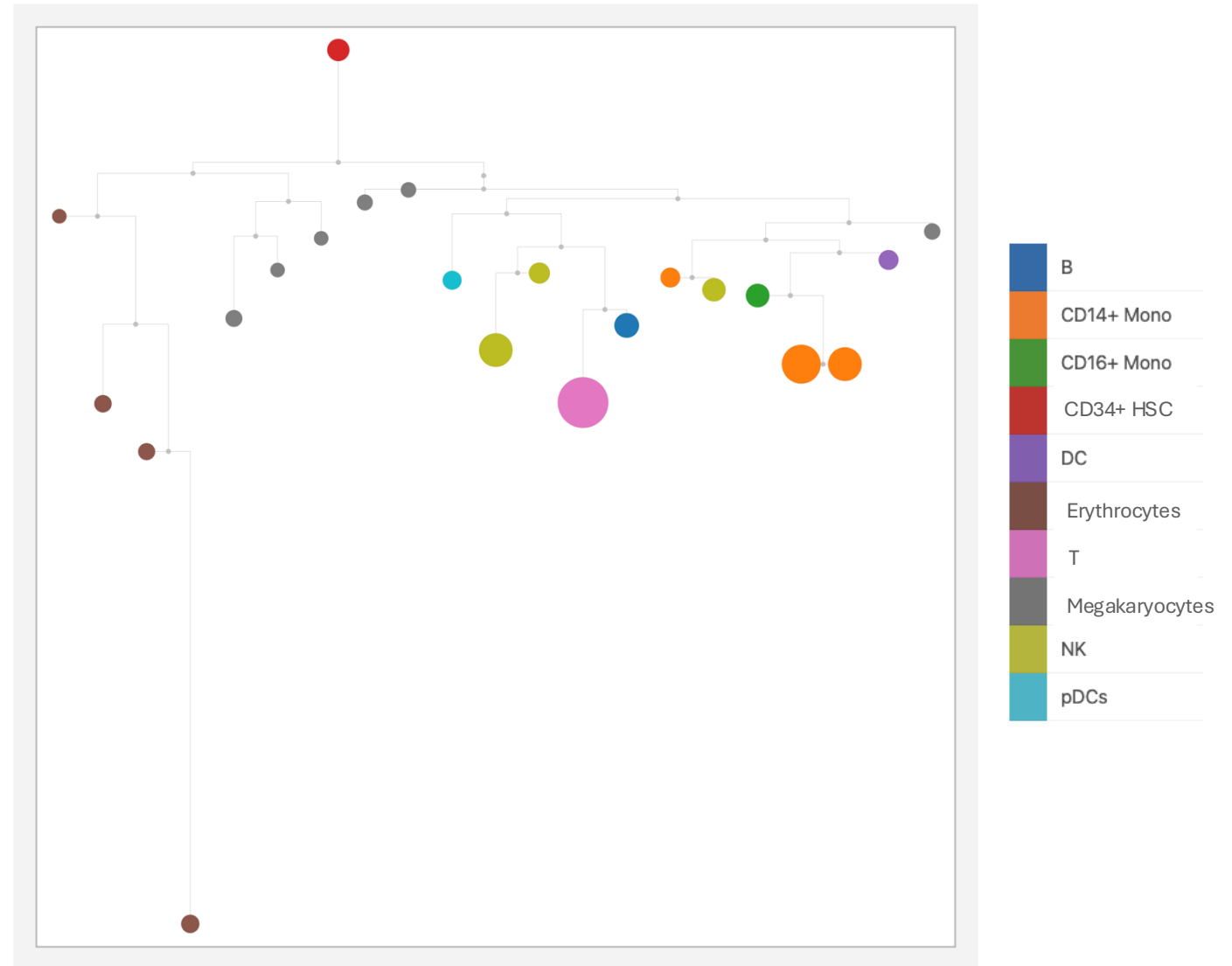


Sarah Morillo

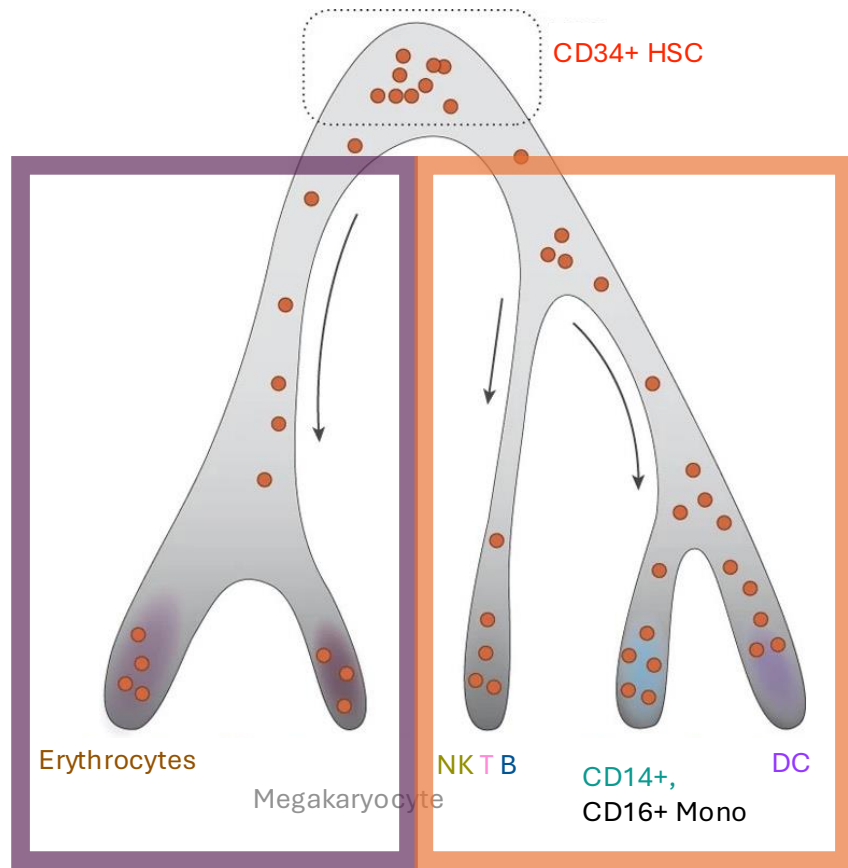
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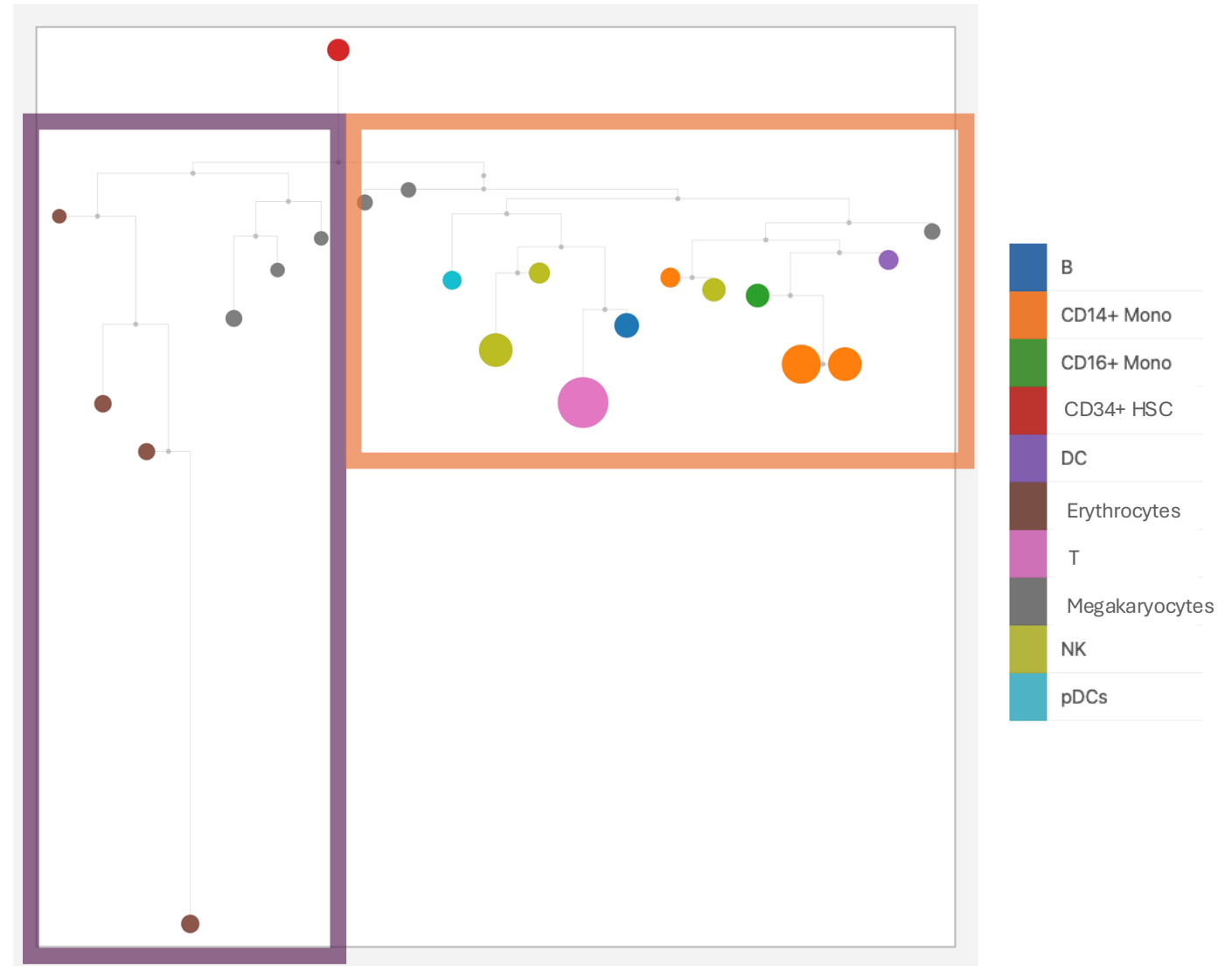
Laurent and Göttgens, *Nature*, 2018



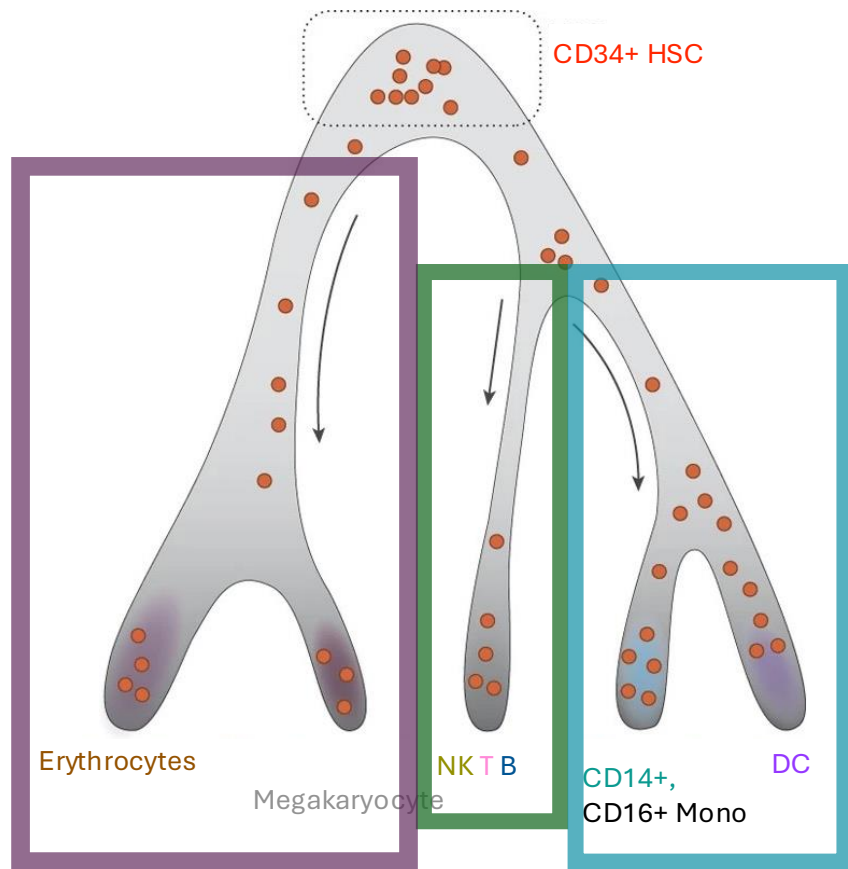
Bonsai reconstructs correct differentiation trajectories on a blood-cell dataset **without tuning any parameter**



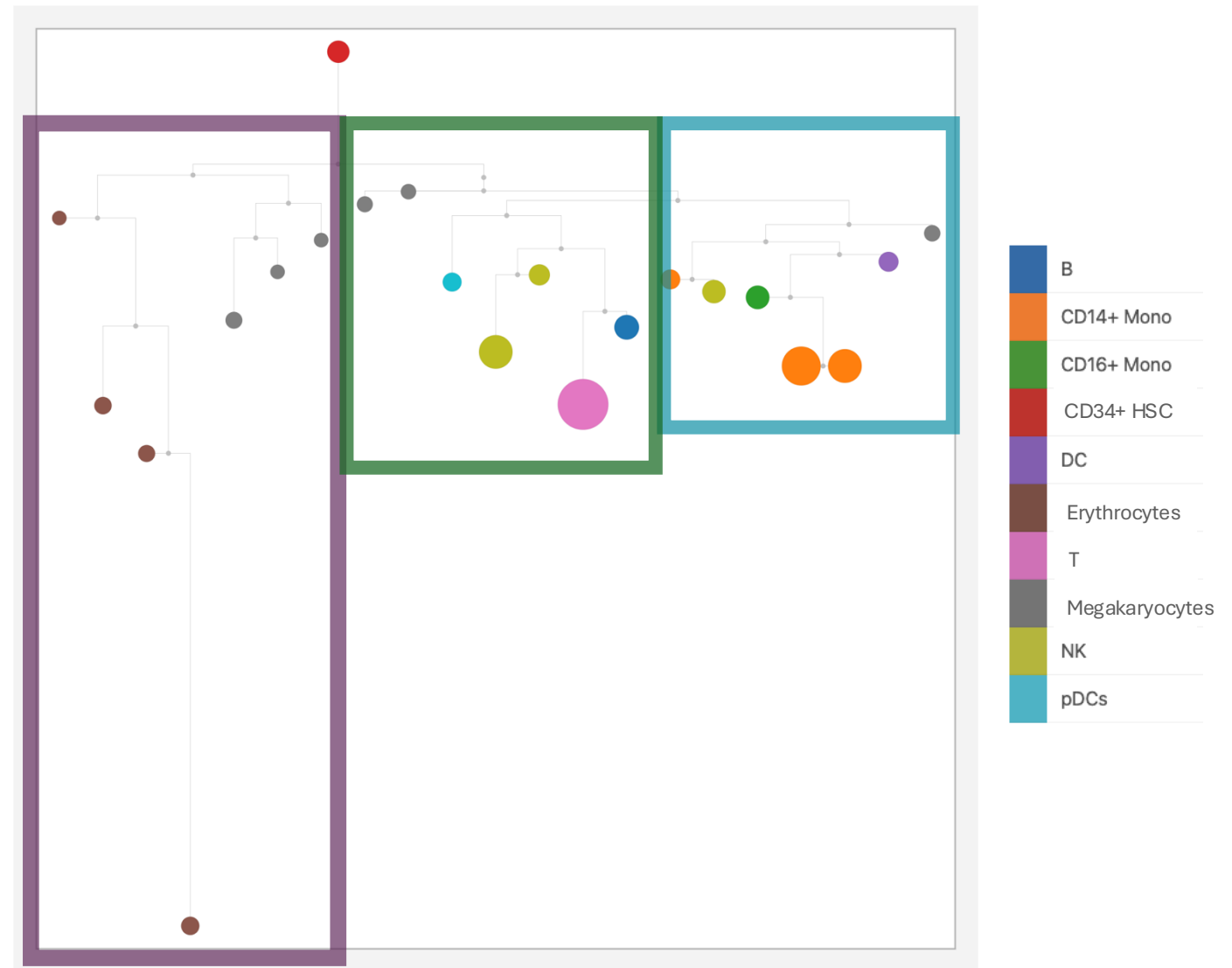
Laurent and Göttgens, *Nature*, 2018



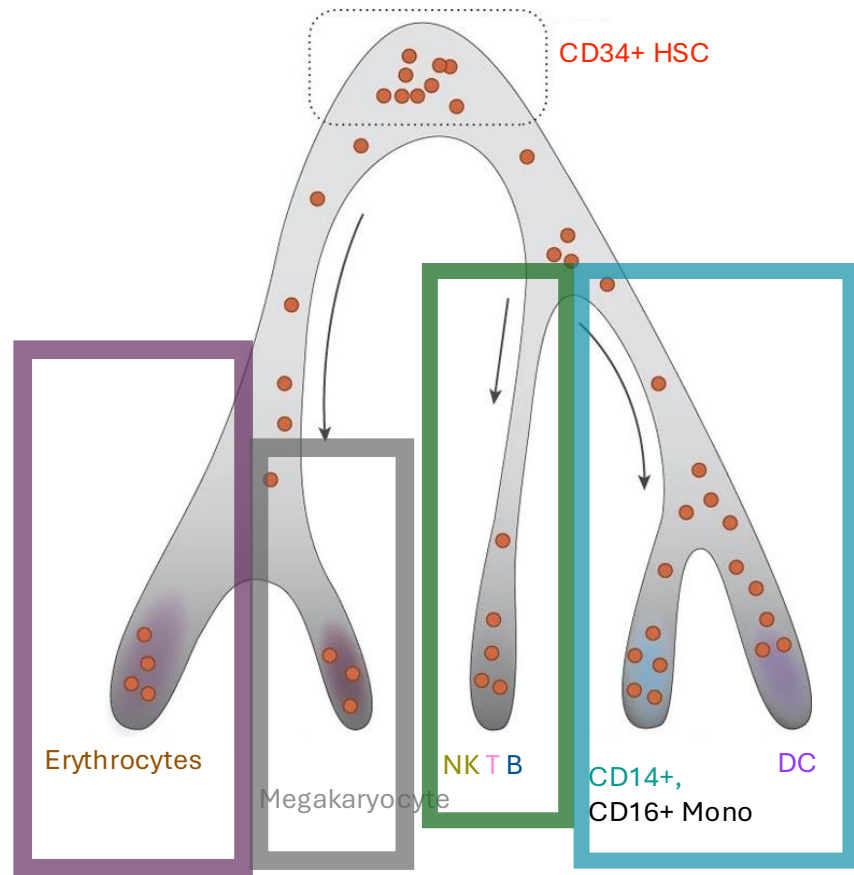
Bonsai reconstructs correct differentiation trajectories on a blood-cell dataset **without tuning any parameter**



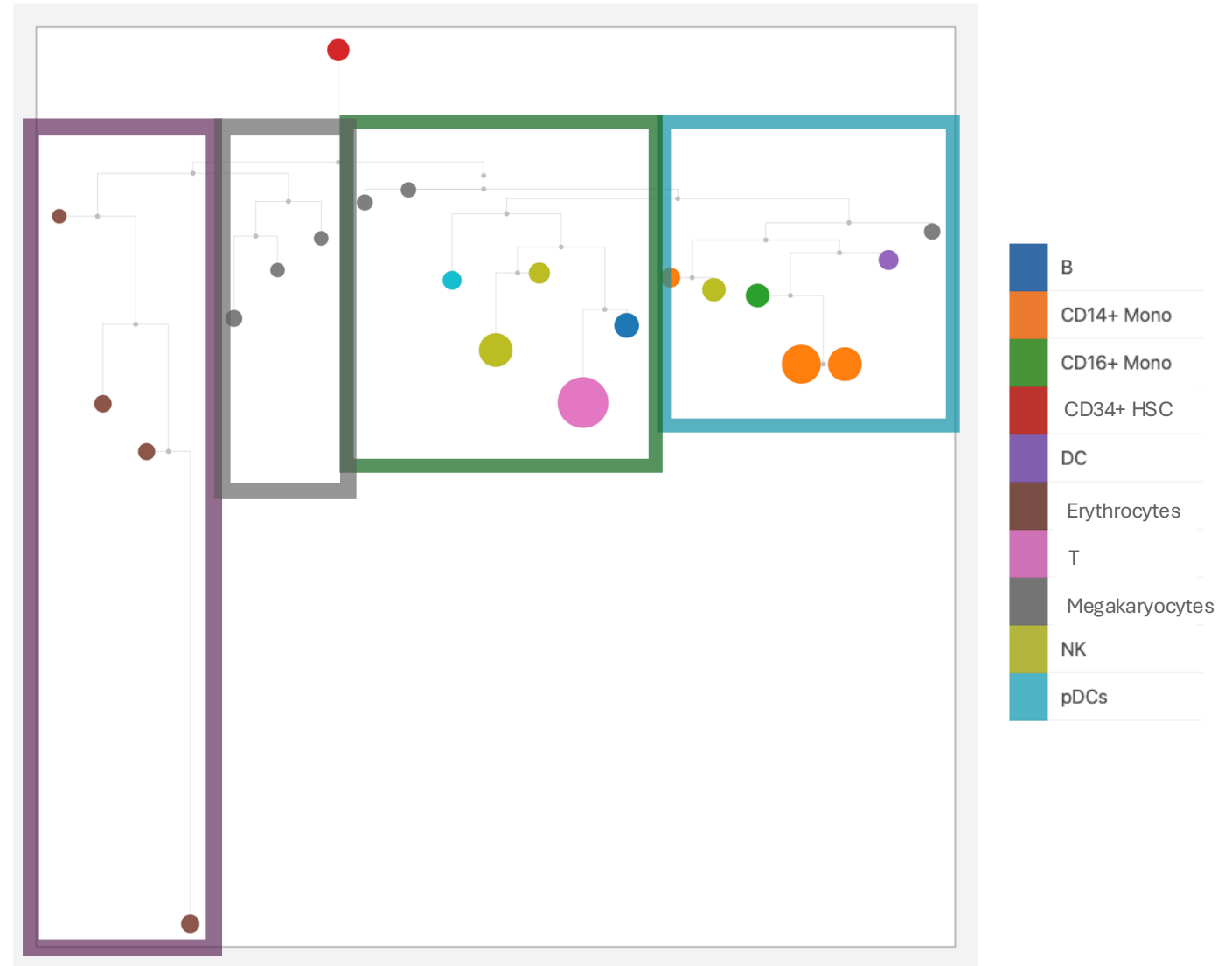
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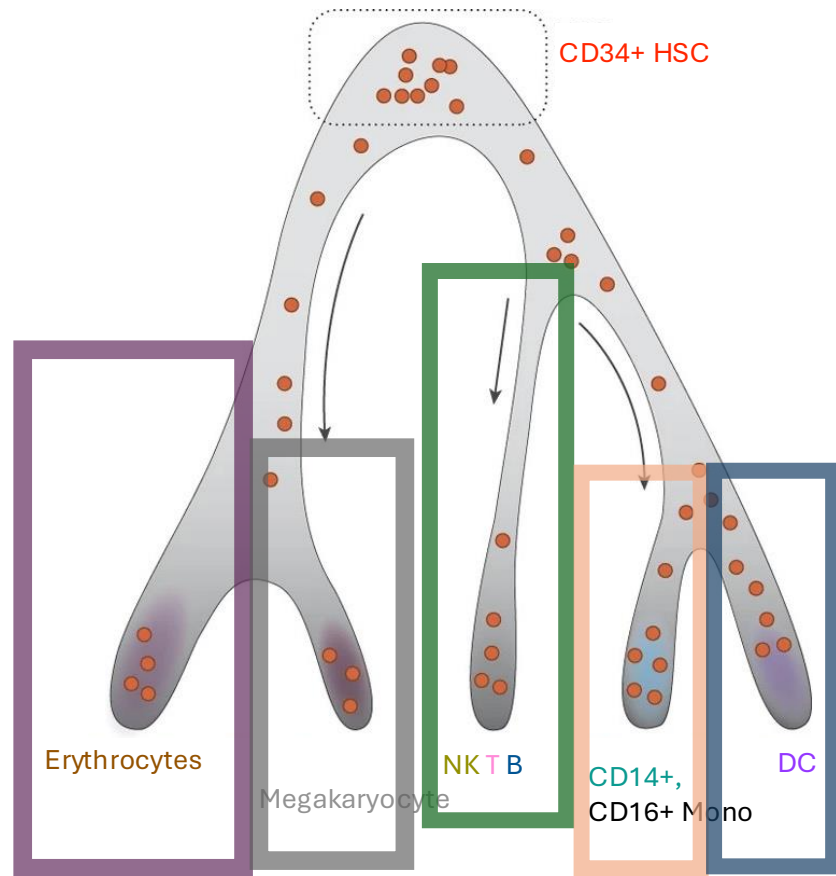
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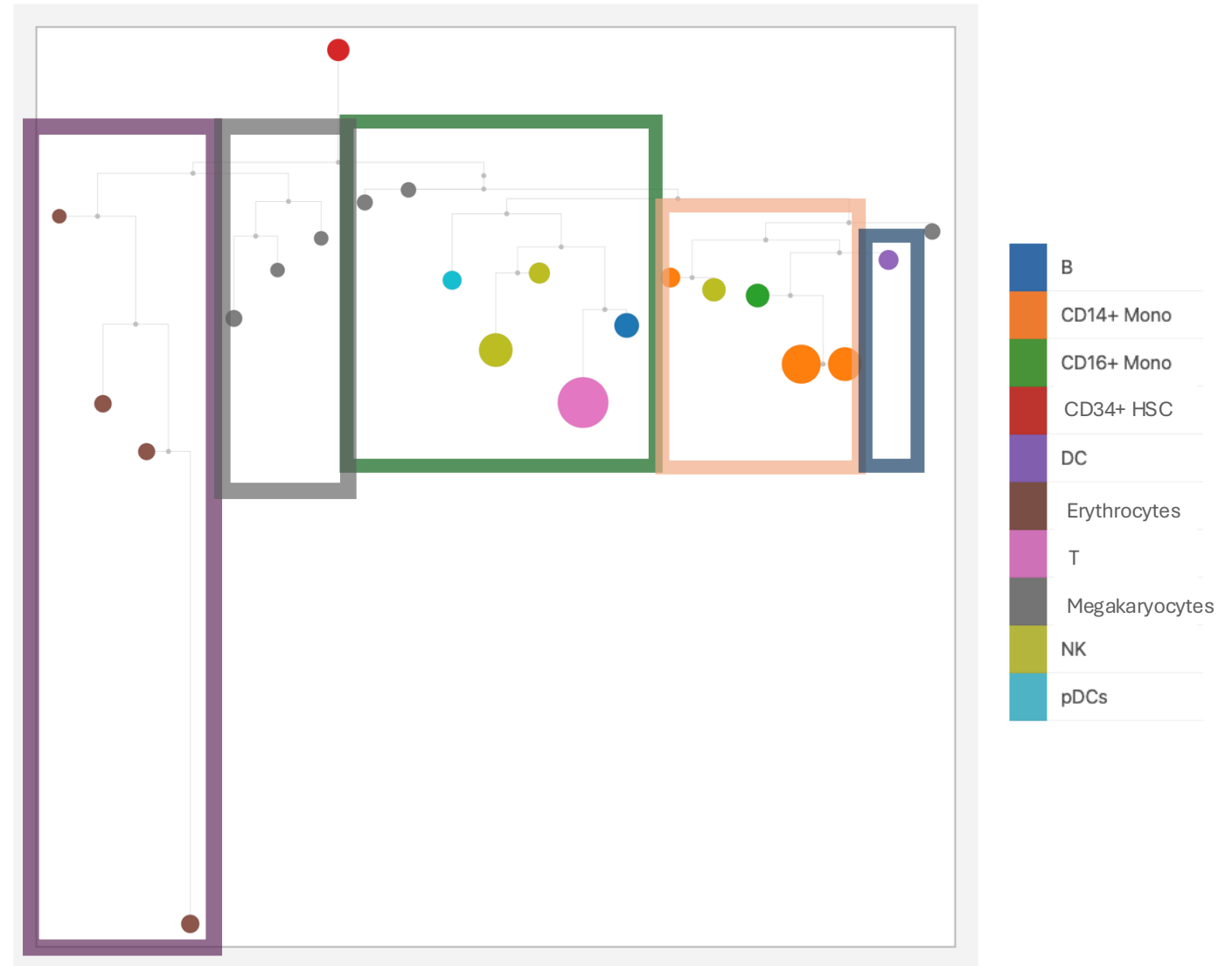
Laurent and Göttgens, *Nature*, 2018



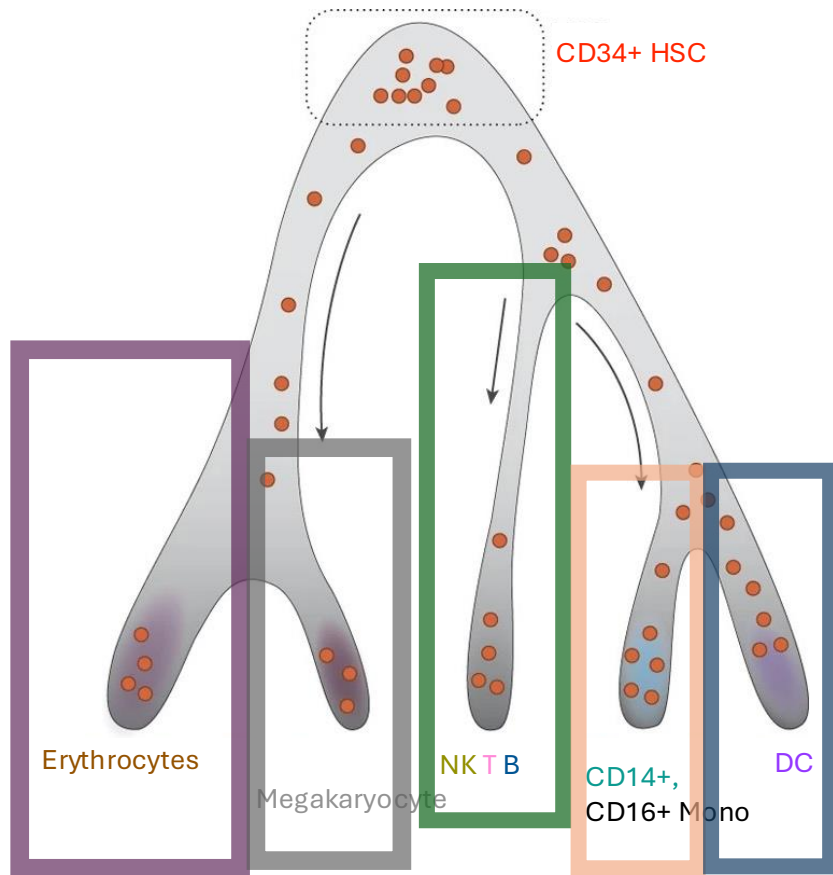
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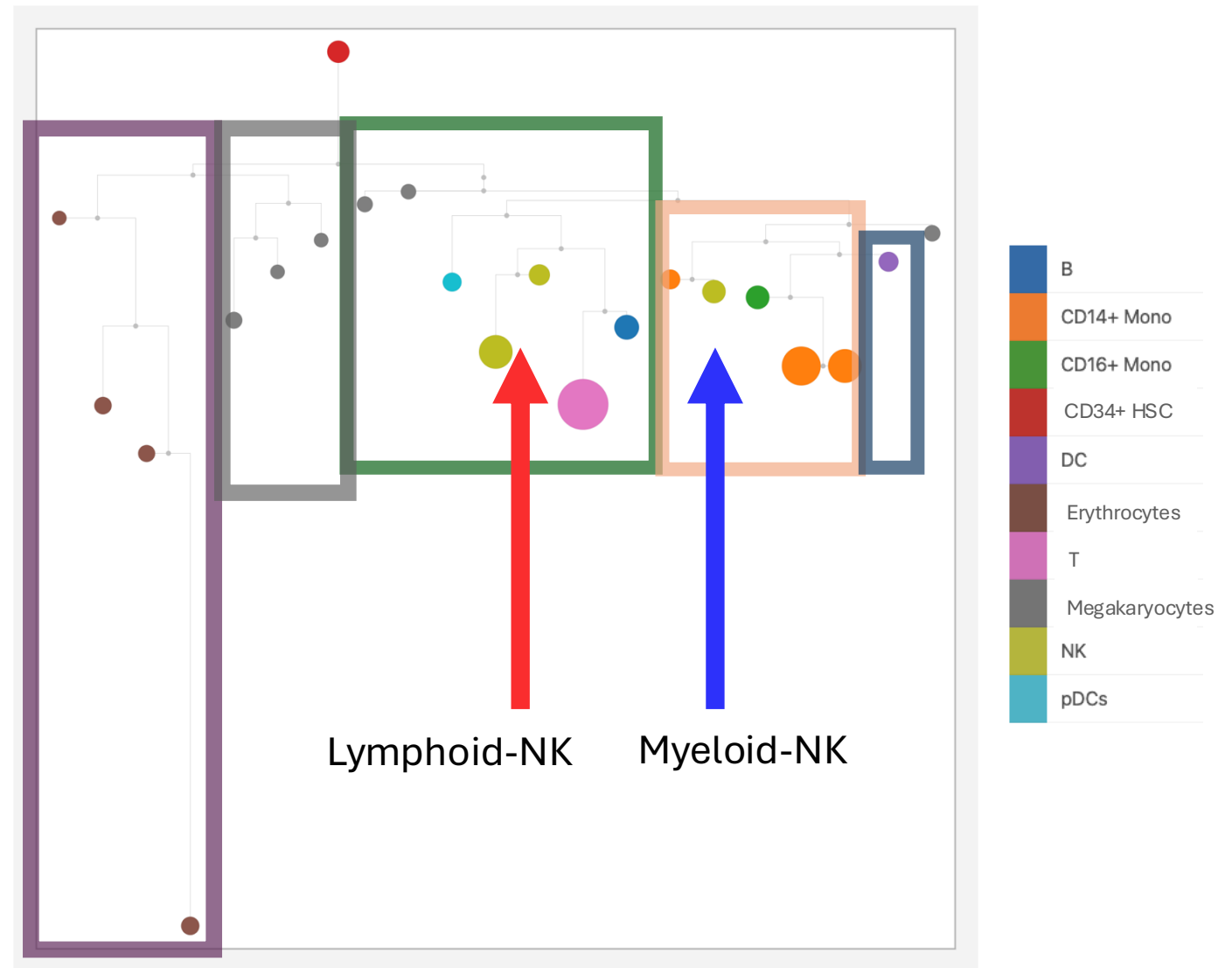
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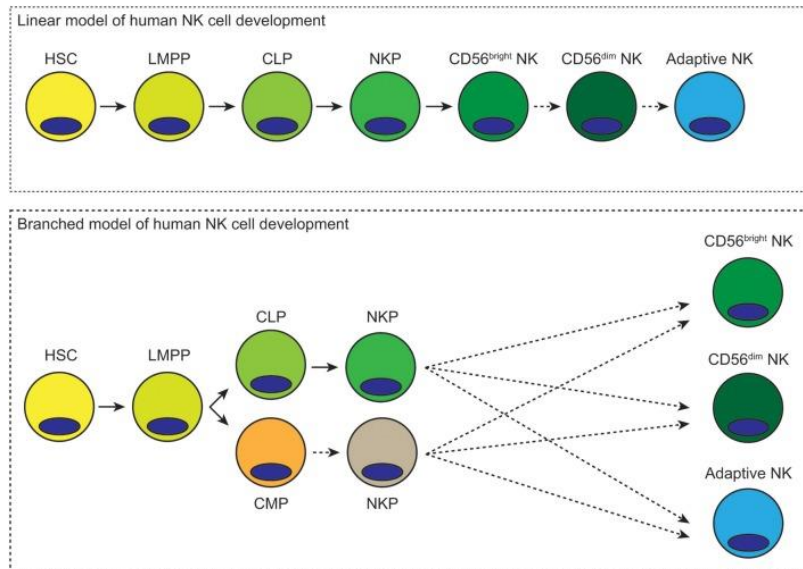
Bonsai finds *in vivo* evidence for myeloid-derived NK cells



Laurent and Götting, *Nature*, 2018

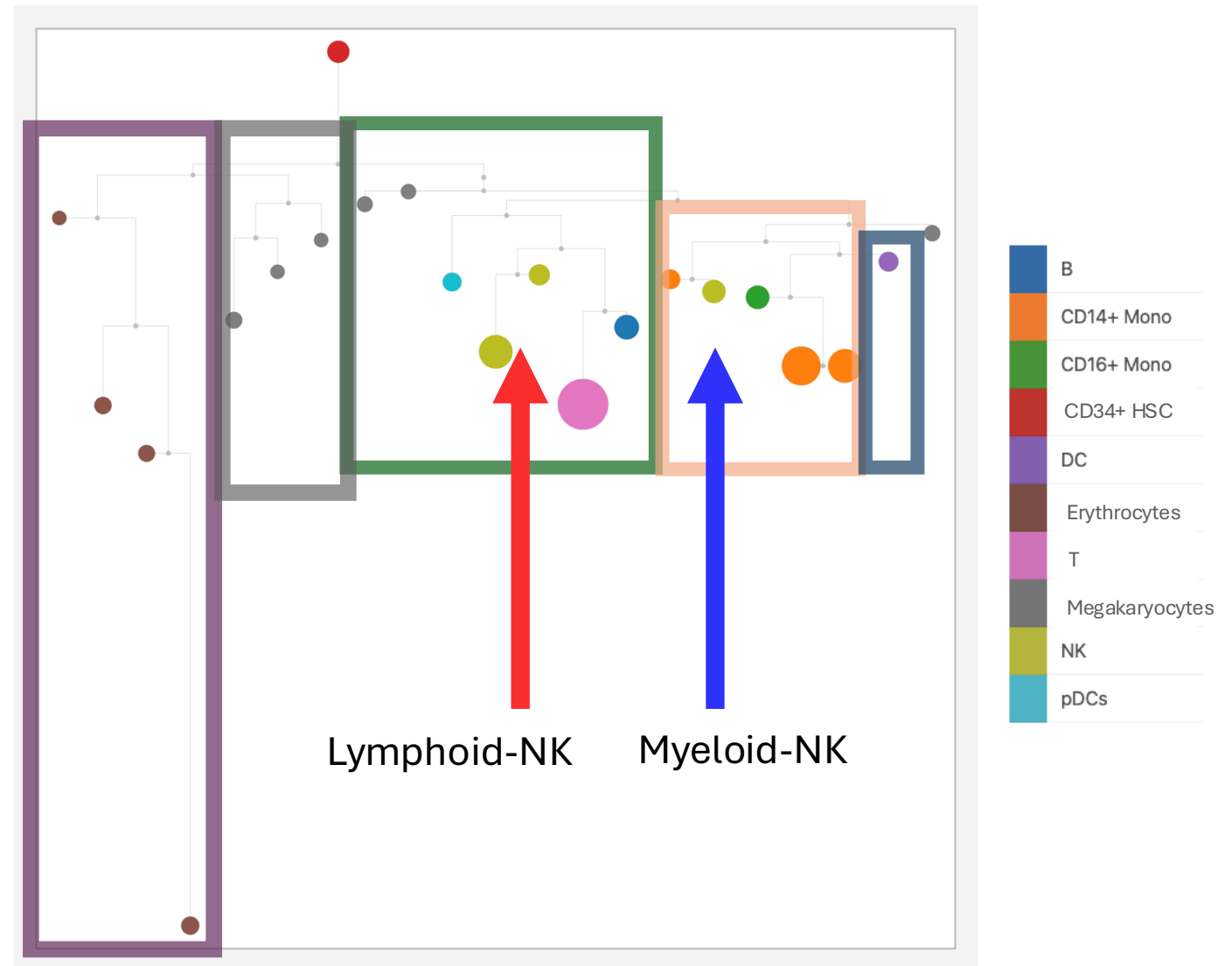


Bonsai finds *in vivo* evidence for myeloid-derived NK cells

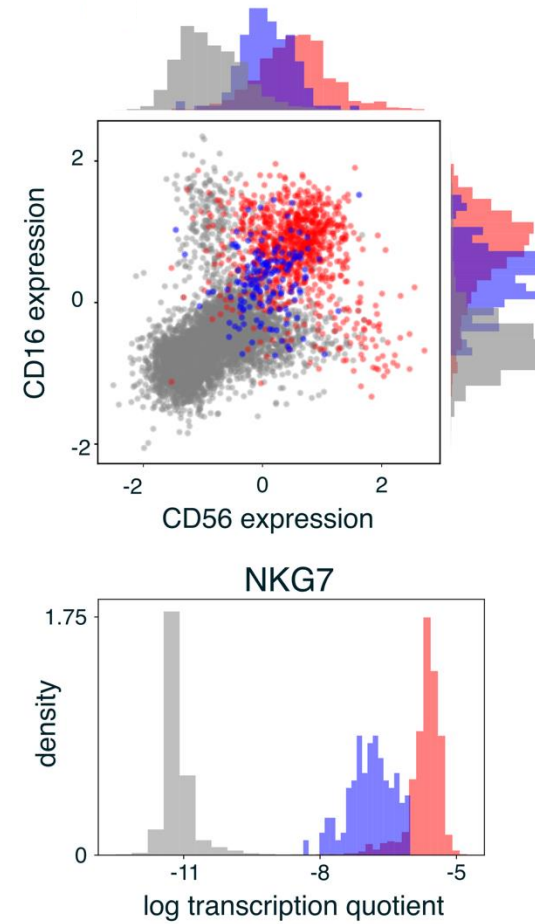
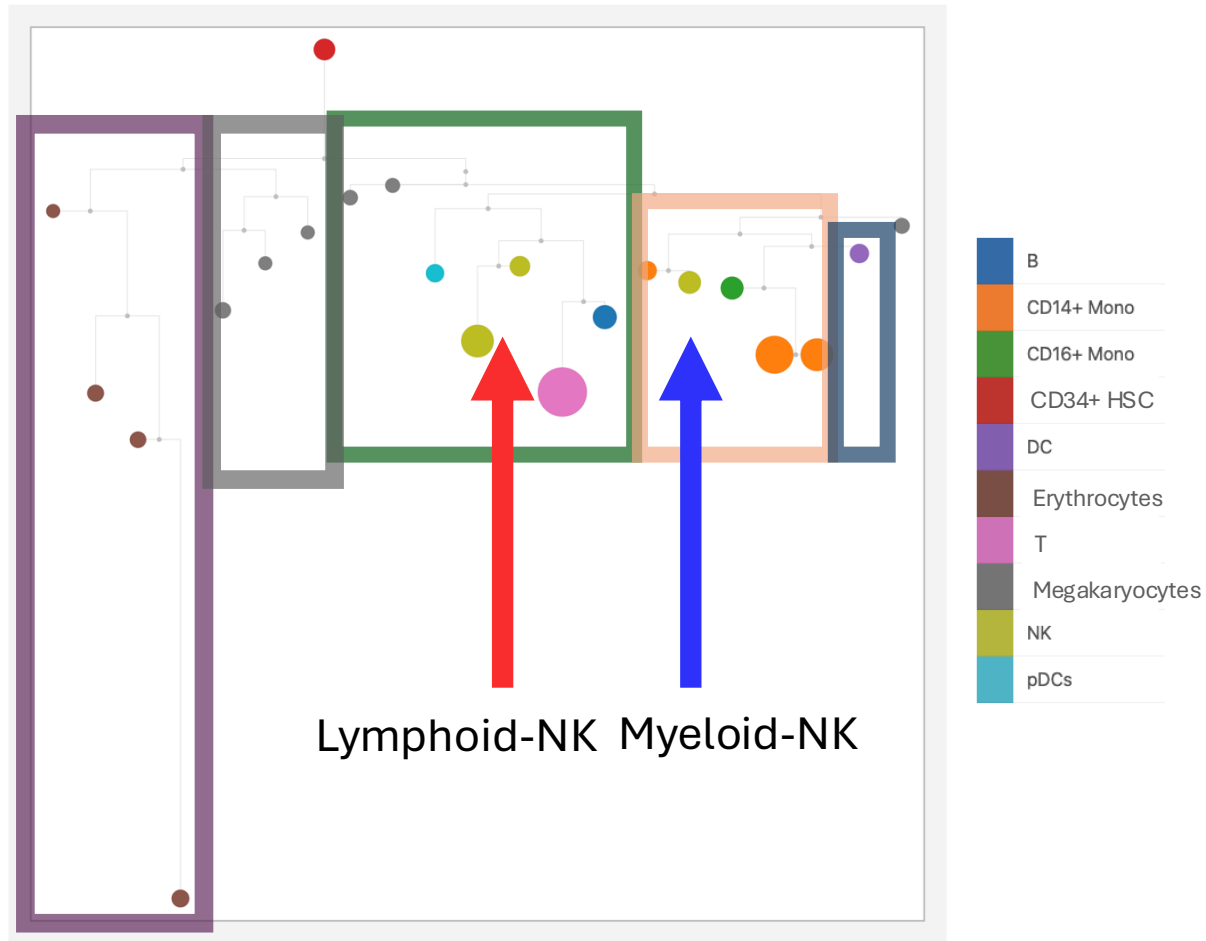


Cichocki et al. (2019)

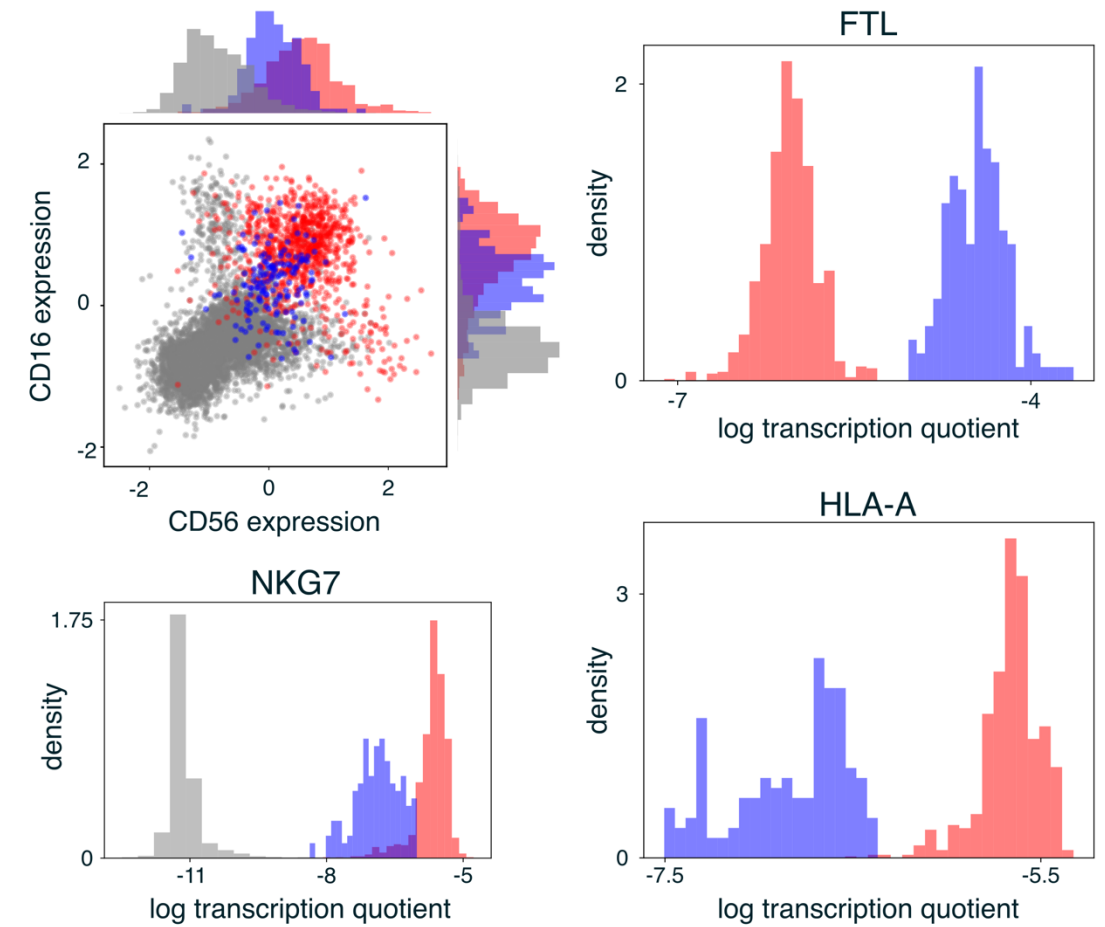
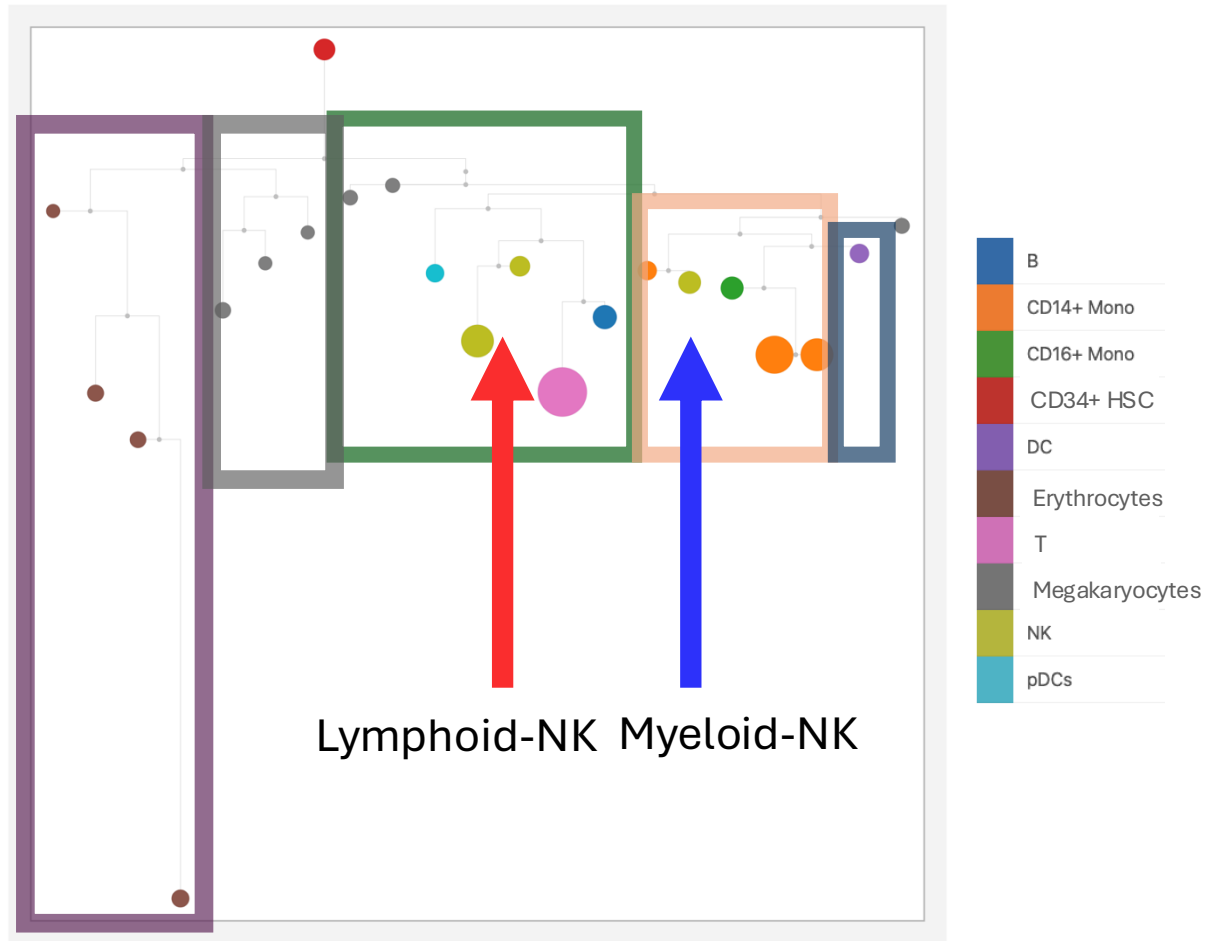
Human NK Cell Development: One Road or Many?



Bonsai finds *in vivo* evidence for myeloid-derived NK cells



Bonsai finds *in vivo* evidence for myeloid-derived NK cells



Today's program

- **14:00 – 14:30:** Challenges in scRNA-seq data analysis
- **14:30 – 15:00:** Our solution: *Sanity* + *Bonsai*
- **15:00 – 16:15:** Hands-on tutorial on running *Bonsai* and *Bonsai-scout*
- **16:20 – 16:50:** From dataset to paper figure
- **17:00 – 18:00:** Q&A + individual dataset analysis

The tree likelihood is independent of the choice of root

Practical note: the likelihood is independent on the choice of root. Because we took a uniform prior over the root position, and because the expression for the likelihood of movement is symmetric in $\mathbf{x}_j$ and $\mathbf{x}_{\pi(j)}$, we have the following identity

$$P(\mathbf{x}_j|\mathbf{x}_{\pi(j)}, t_j)P(\mathbf{x}_{\pi(j)}) = P(\mathbf{x}_{\pi(j)}|\mathbf{x}_j, t_j)P(\mathbf{x}_j). \quad (\text{S14})$$

As a result, we can always designate another node as the root without affecting the tree likelihood. We will use this freedom to choose the root node in several derivations and computational implementations.

Marker genes

Let $\bar{C}_1$ and C_2 denote two clades of cells on the tree. We define marker genes as genes that either maximize or minimize the probability that, when picking a random pair of cells, $c_1 \in C_1$ and $c_2 \in C_2$, the gene expression x_{gc_1} is higher than x_{gc_2} . Dropping the gene-indices, we thus want to calculate for each gene:

$$M = \frac{1}{|C_1||C_2|} \sum_{c_1 \in C_1} \sum_{c_2 \in C_2} P(x_{c_1} > x_{c_2}). \quad (9)$$

To calculate this, we use the posteriors over the positions of the cell-associated leaf-nodes that *Bonsai* provides. Specifically, for cell c_1 , the posterior $P(x_{c_1}|D, T, \mathbf{t})$ is a Gaussian distribution with mean $\bar{\mu}_{c_1}$ and variance $\bar{\sigma}_{c_1}^2$, and similarly for cell c_2 . To obtain the probability $P(x_{c_1} > x_{c_2})$, we note that the probability distribution of the difference, $x_{c_1} - x_{c_2}$ is also a Gaussian with mean $\bar{\mu}_{c_1} - \bar{\mu}_{c_2}$ and variance $\bar{\sigma}_{c_1}^2 + \bar{\sigma}_{c_2}^2$. Therefore, we can express the desired probability in terms of the cumulative distribution of these Gaussians, which can be efficiently calculated using the error-functions:

$$M = \frac{1}{|C_1||C_2|} \sum_{\substack{c_1 \in C_1 \\ c_2 \in C_2}} \left(\frac{1}{2} + \frac{1}{2} \operatorname{erf} \left(\frac{(\bar{\mu}_{c_1} - \bar{\mu}_{c_2})}{\sqrt{2(\bar{\sigma}_{c_1}^2 + \bar{\sigma}_{c_2}^2)}} \right) \right). \quad (10)$$

Marker genes are those for which M is close to 1 or 0, corresponding to high probability for the cell to belong to group C_1 and C_2 , respectively. We provide a supplementary tool with the *Bonsai* code that takes a file that specifies two groups of cells and calculates the marker scores M for all genes. This supplementary tool can be found in the *Bonsai* code repository under [downstream_analyses/calc_marker_genes.py](#).

Unsupervised clustering

In the unsupervised method, branches are iteratively cut so as to greedily minimize the sum of pairwise distances between all leaves in the created subtrees. More specifically, for any tree, we can calculate the sum of pairwise distances between all leaves along the branches of the tree. Note that this sum differs from the sum of all branch lengths, because branches in the middle of the tree are traversed by many more leaf-to-leaf paths than branches near the leaves. Starting from the original tree, we create two subtrees by cutting the branch that minimizes the combined sum of pairwise distances between leaves in the two resulting subtrees. Given these new trees, we can now cut another branch to again minimize the pairwise distances in the three resulting subtrees, and so on. To create n clusters, one repeats this cutting procedure $n - 1$ times.

Supervised clustering

In the supervised method, we search for a way of cutting the tree into subtrees such that the normalized mutual information (NMI) between subtree membership and the labels of the user-provided annotation is maximized. Specifically, we search over all possible subtree-sets by combining

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